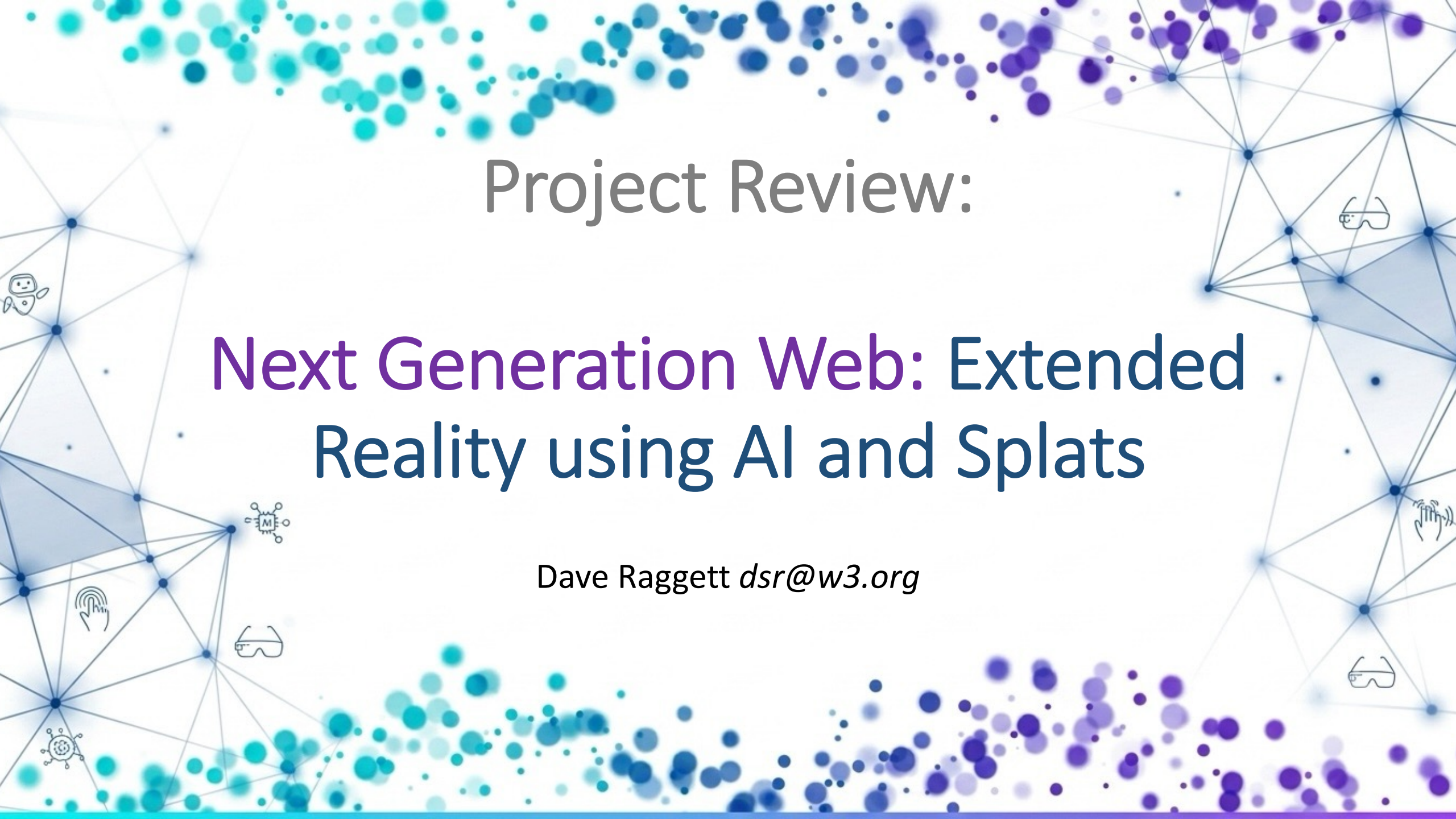




Project Review:

# Next Generation Web: Extended Reality using AI and Splats

Dave Raggett *[dsr@w3.org](mailto:dsr@w3.org)*

# Talk Outline

- Tomorrow's Web
  - Free choice in how you interact and appear in the immersive web
- Technical Challenges
  - Architecture for Immersive Presence
  - Farewell To Triangles
  - Pointillism and Gaussian Splats
  - Liquid Neural Networks
  - Pre-training on large corpora
  - Personalization in the browser
- Related standards efforts
- A Web of agents & digital twins
- Democratizing AI development with WebNNM
- Community driven development of the Immersive Web
- Where next for AI and Knowledge-based Systems
- Opportunities for W3C

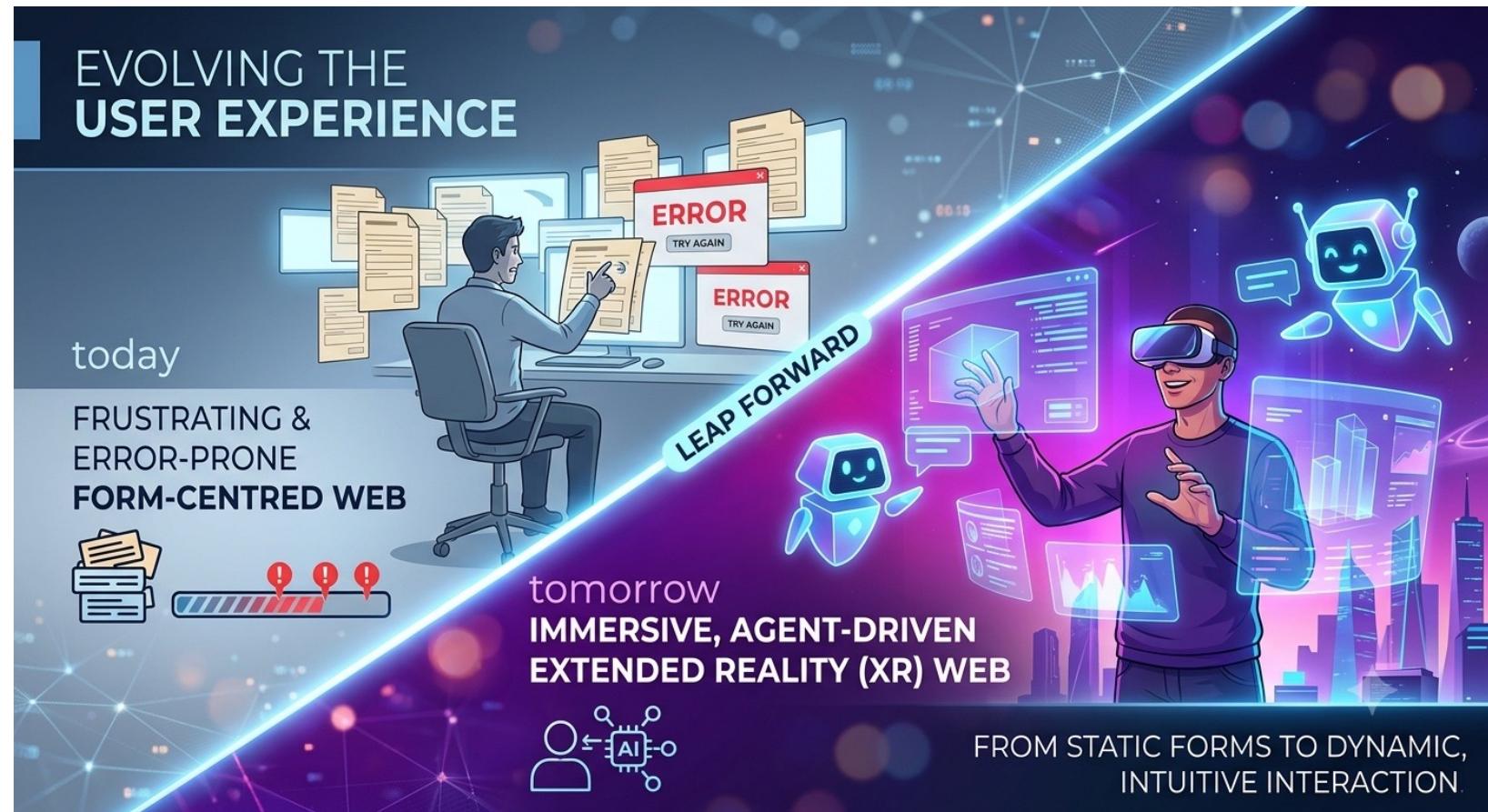
# Tomorrow's Web

# Tomorrow's Web

Today's Web focuses on form filling which is frustrating and error prone

*Partly my fault, as I introduced forms into web pages in the early nineties!*

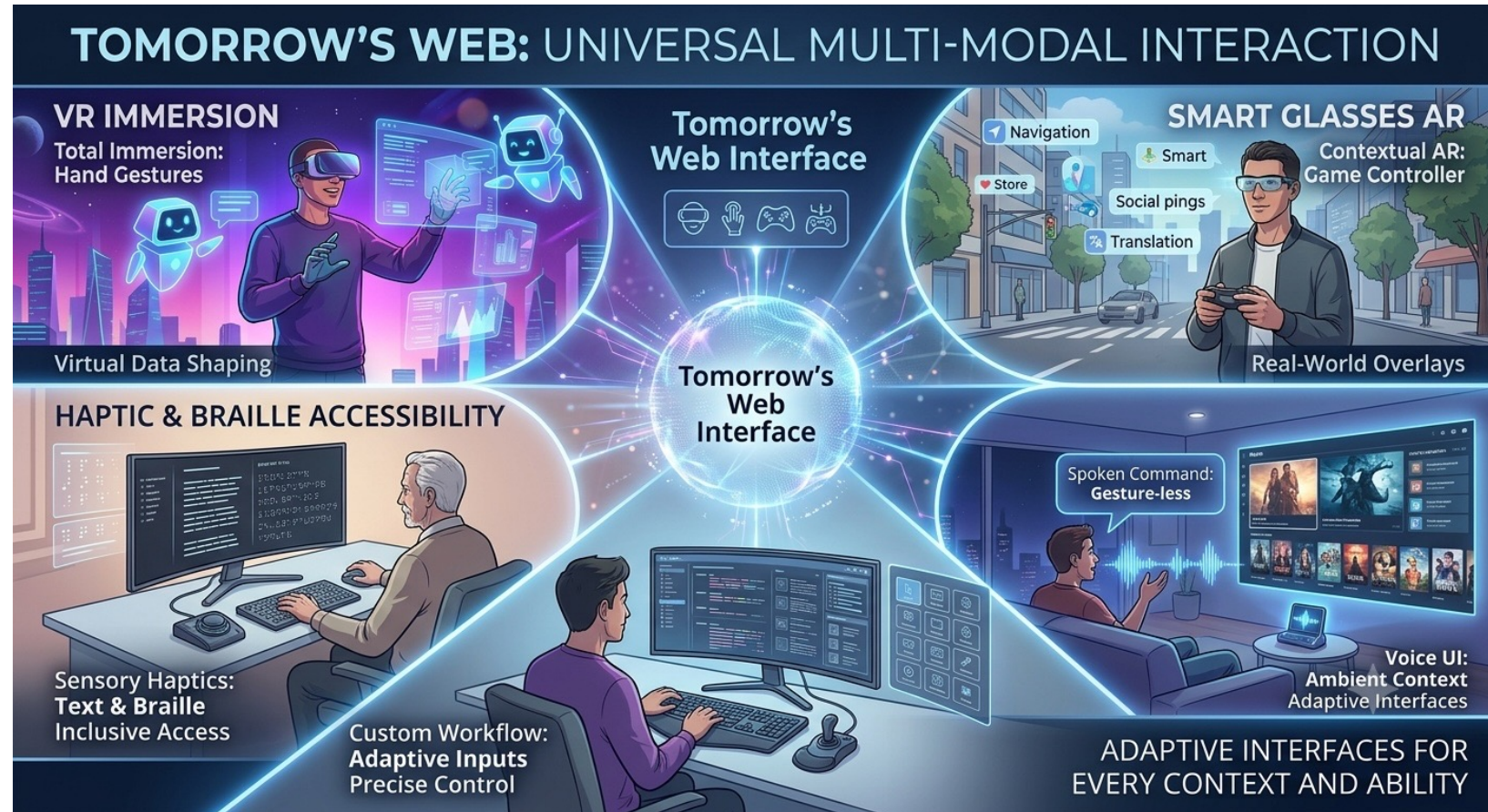
Tomorrow's Web will instead focus on AI agents and immersive experiences in extended or virtual reality



# Personal Freedom in How You Interact

People will be able interact with tomorrow's web in many different ways according to their physical abilities, personal preferences and the social context they find themselves in.

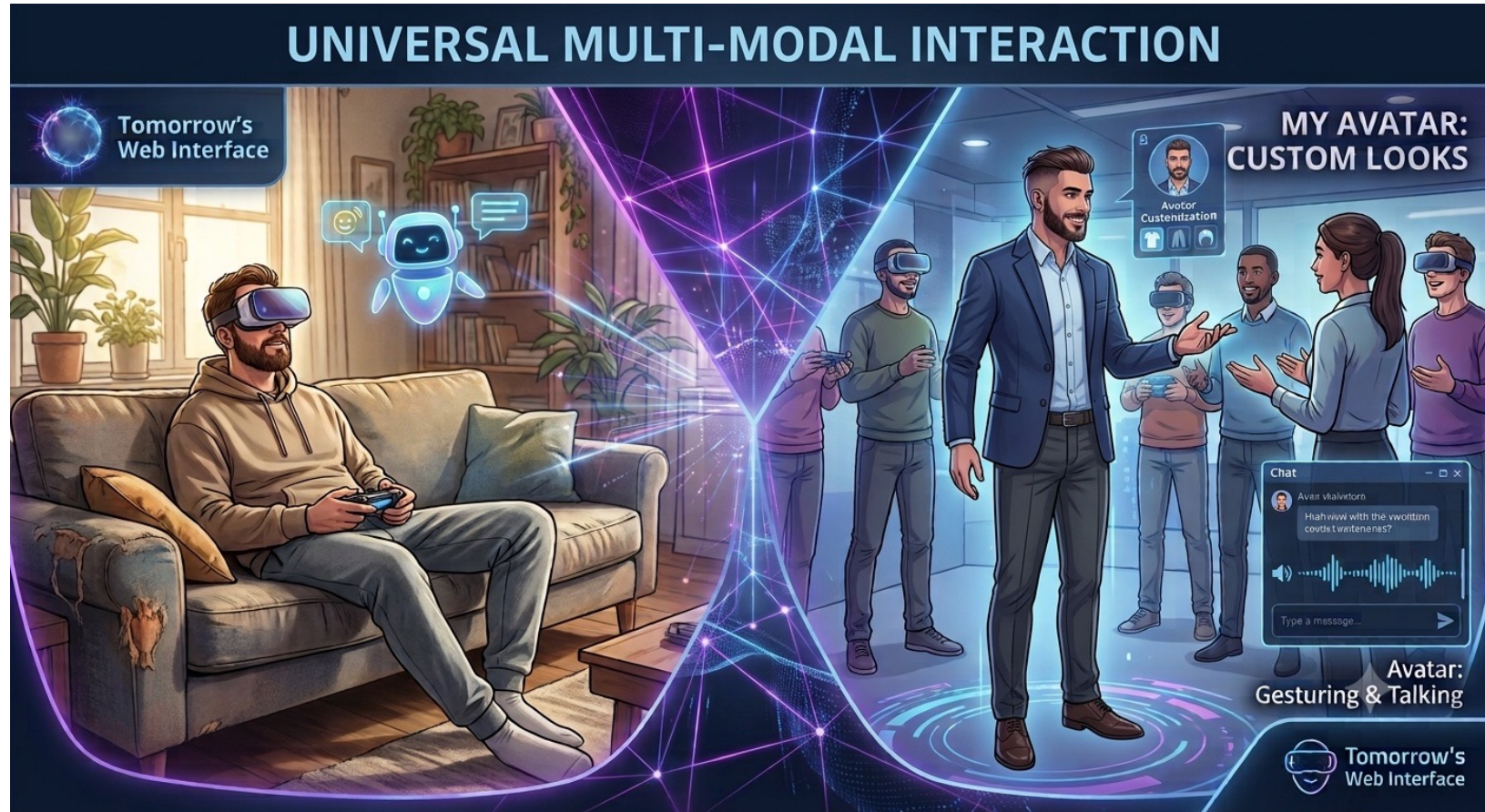
Some people will use VR headsets, others smart glasses, or even regular large monitors, along with games controllers, haptic feedback, regular keyboards, hand gestures and spoken commands.



# Personal Freedom in How You Appear

You may be slouched in your living room, but your avatar is walking, talking, gesturing, and looking at people.

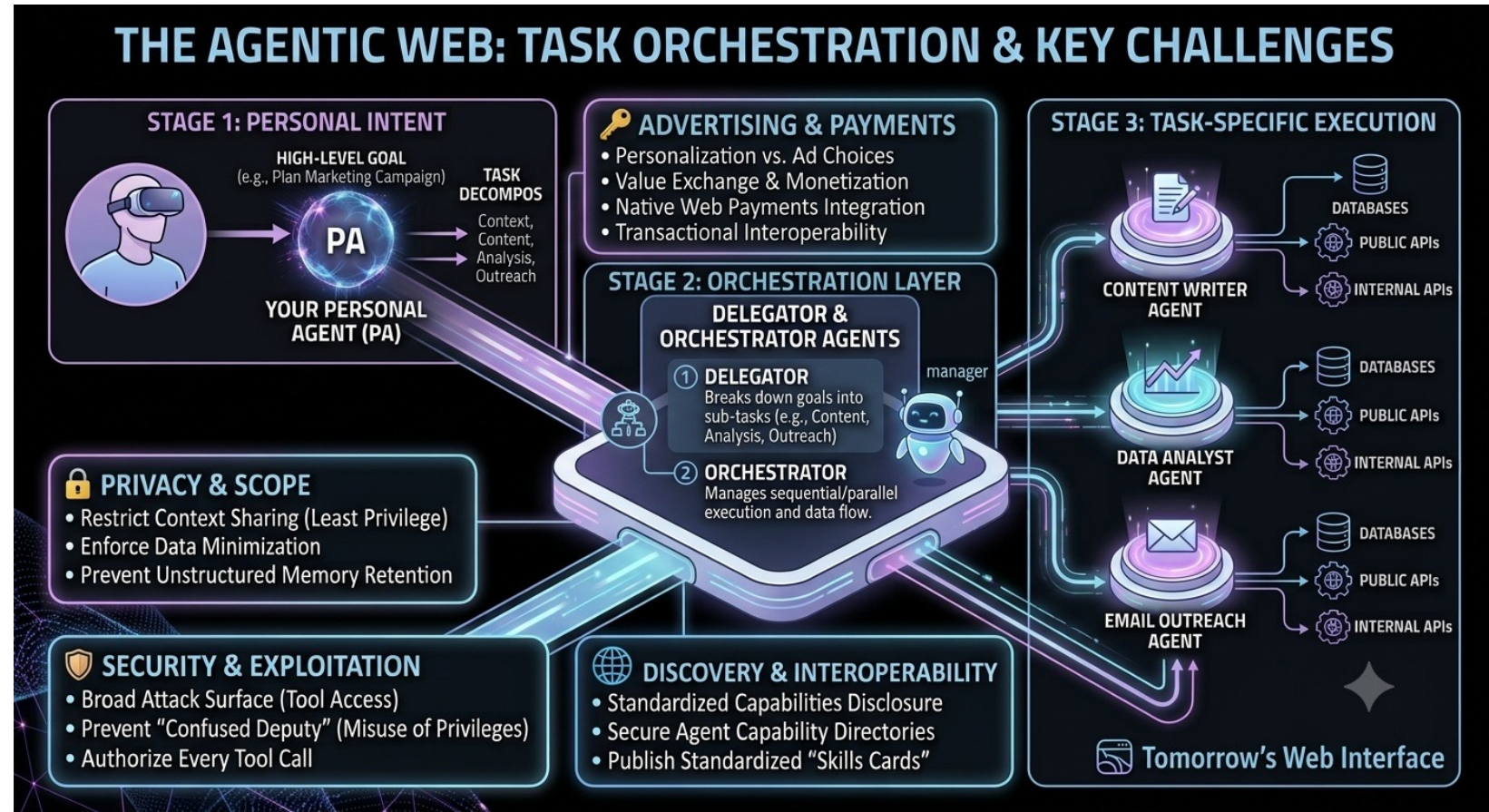
You choose your avatar's appearance to suit the occasion: formal or fun.



# A Web of Agents

Your personal agent talks to other agents, which delegate tasks to further agents, which in turn access tools.

Lots of challenges relating to privacy, trust, discovery, advertising, security and payments.

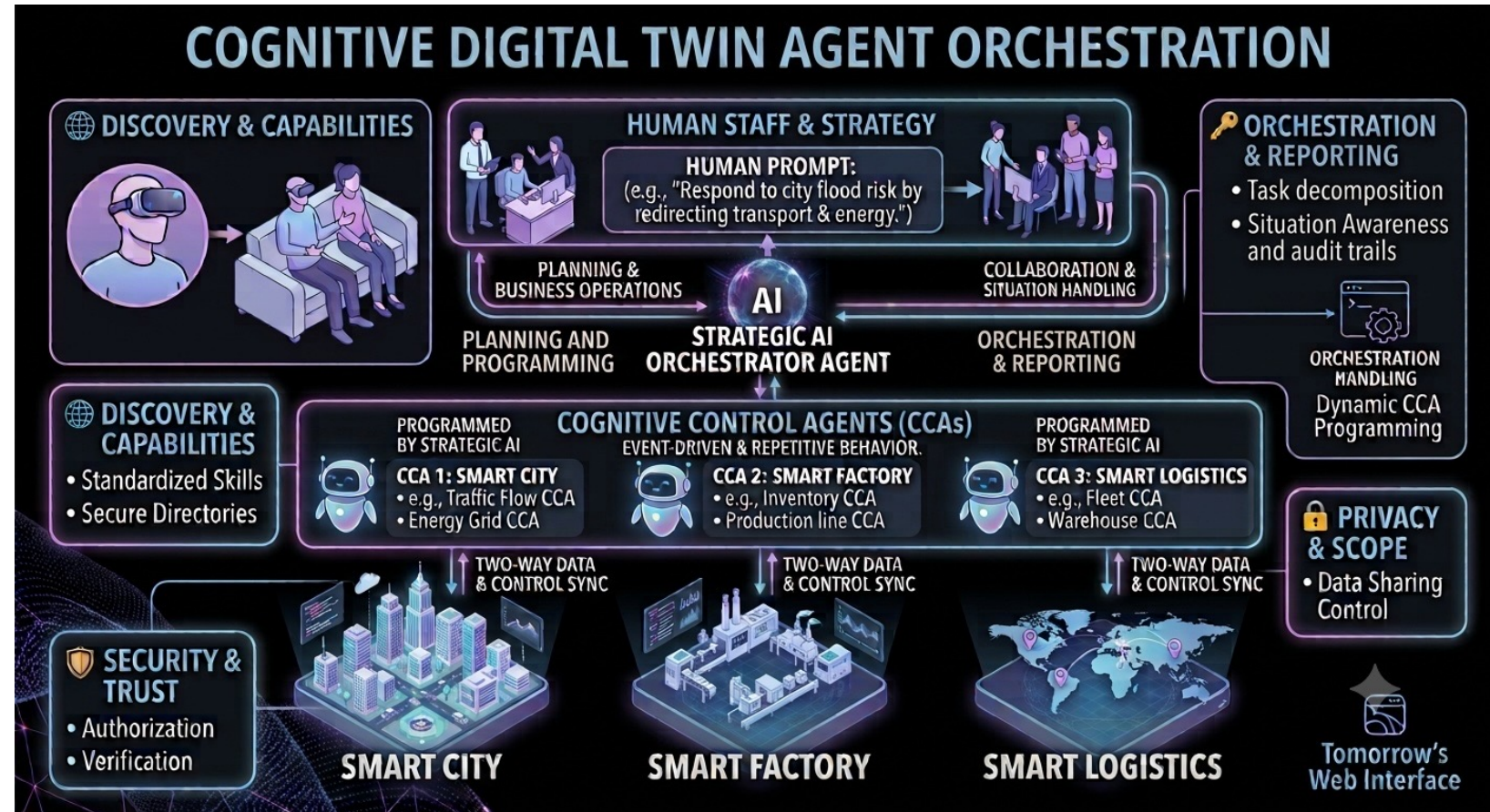


*Will advertising to agents replace direct advertising to consumers as agents intermediate services?*

# Digital Twins and Cognitive Control

A digital twin is a dynamic, real-time virtual model of a physical object, system, or process, using real-world sensors and actuators.

It allows you to simulate performance, predict failures, and test changes in a risk-free digital environment before applying them in reality.



*Use of AI agents to program simpler cognitive control agents for event-driven repetitive behaviours*

# Immersive Scenes and Stage Directions

- In a play, the director interprets the scene descriptions and stage directions, deciding on the concept, mood, era, and visual direction
- Something similar is needed for scenes in virtual reality, where we can exploit generative AI for the role of set and costume designers, starting from a text prompt for the scene description and context
- In place of actors, we have human users and AI agents



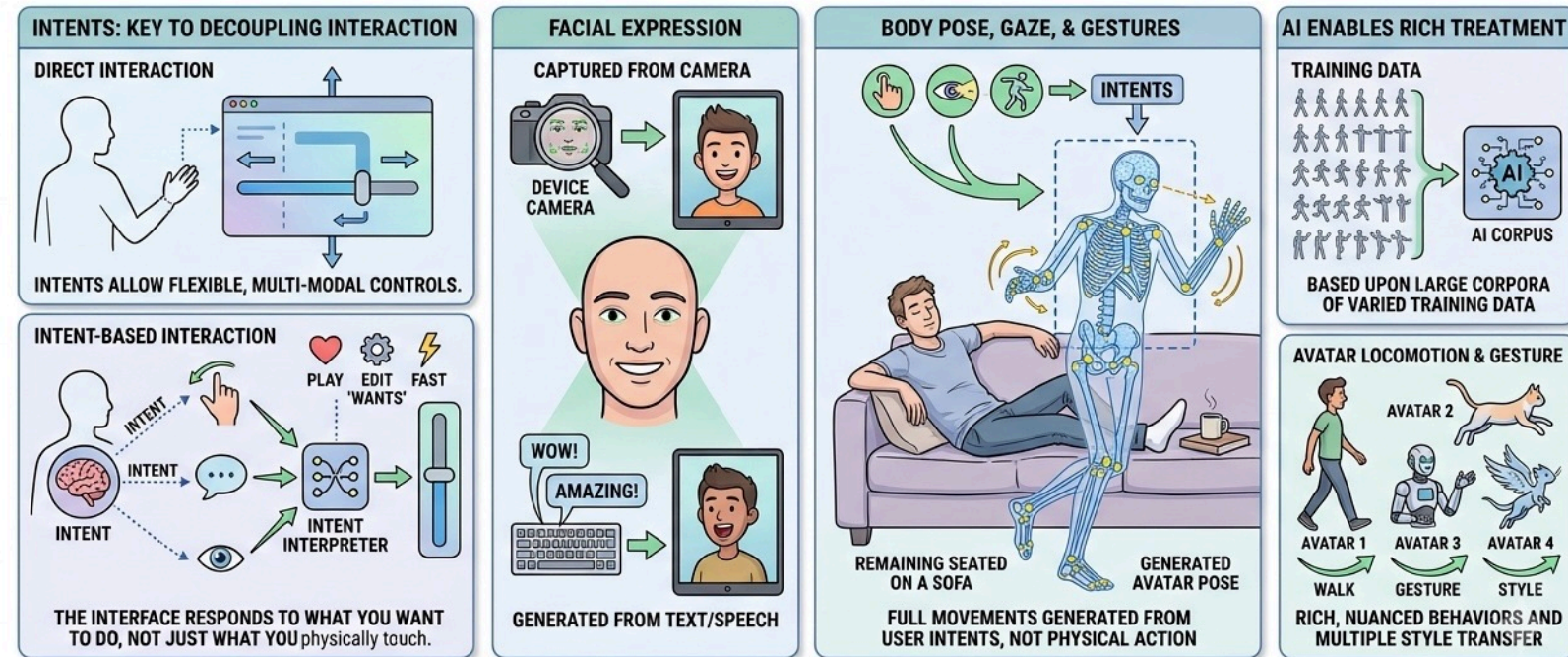
**[SCENE:** A miniature woodland glade set for a small paper-theater production. The backdrop is a painted watercolor vista of green hills, a forest, and scattered flowers under a cloudy sky. The stage itself is constructed of stained wood, complete with tiny, hand-painted wooden footlights and small red curtains drawn back to reveal the scene. The entire production is rendered in a charming, traditional storybook style.]

**[PRESENT:** AN EXPLORER, A TRAVELER, and A STANDING DEER.]

# Intent-Driven Behavior

- **Intents** are the key to freeing users to choose how they interact with scenes to direct their chosen avatar's behavior
- Your facial expression may be captured by your device's camera, or it could be generated to match what you are saying or typing
- Body pose, movements, gaze direction and gestures are **intent generated** as you remain seated on the sofa
- AI enables rich treatment of how avatars move\* based upon training across large corpora, with individual traits rather than uniformity
- Cognitive mapping from high-level to low-level intents, e.g. "open the door" entails planning a route to the door around any obstacles, moving a hand to the door handle, then opening the door

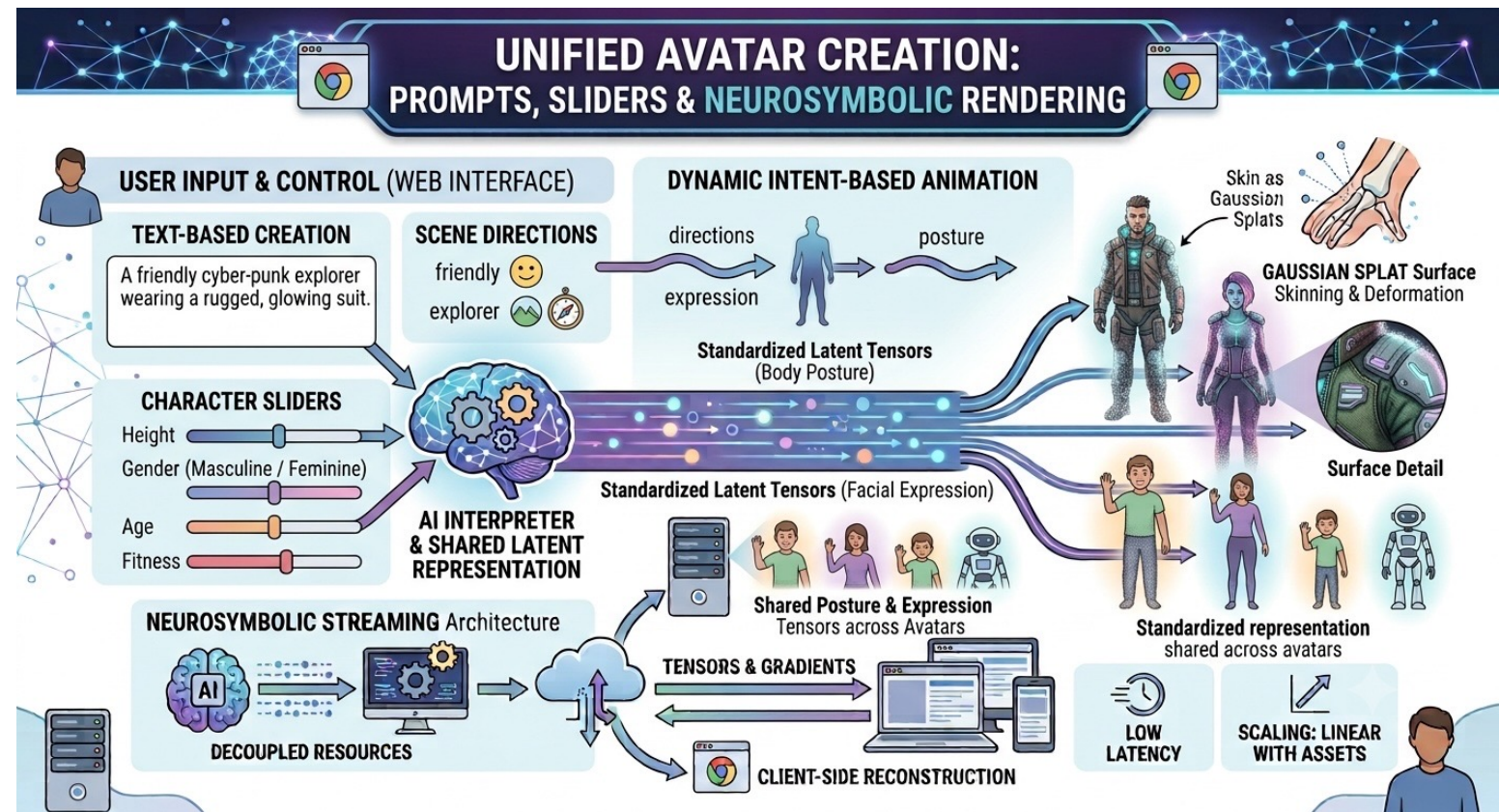
## EXPLORING DIGITAL INTENT AND EXPRESSION



\* Gesturing, standing, sitting, walking, running, dancing, etc.

# Creating Your Custom Avatar

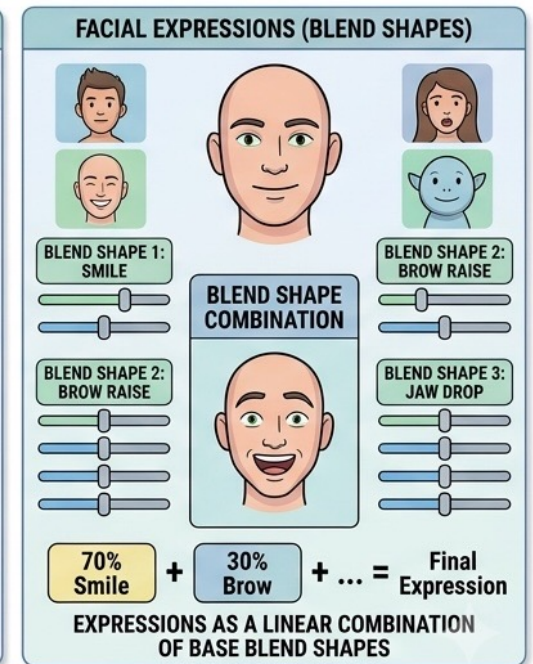
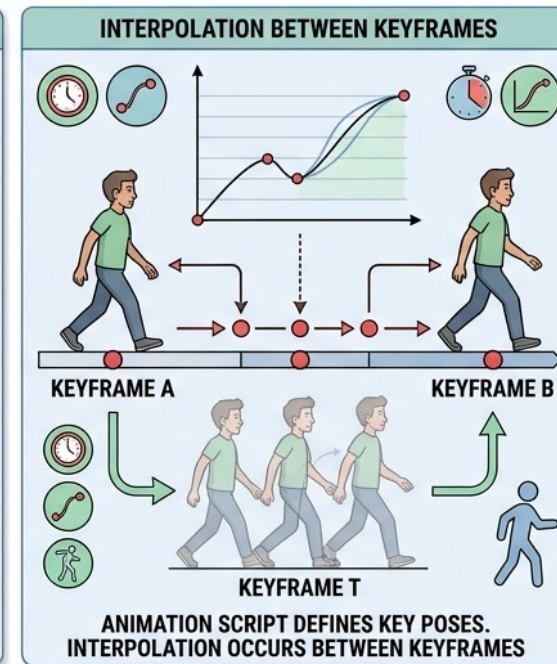
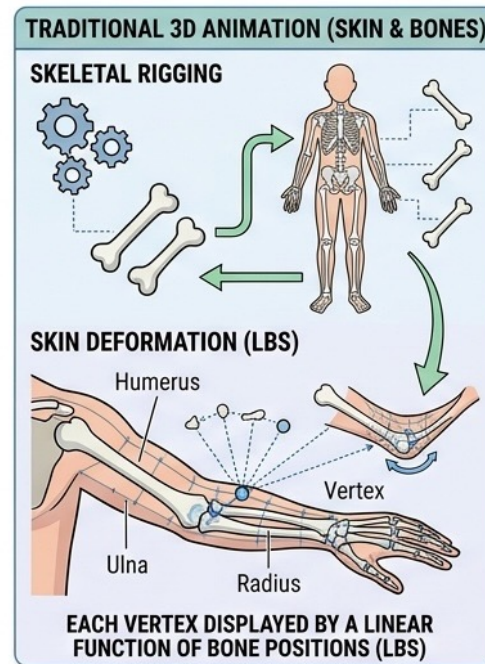
- Users can simply create their own custom avatars
- Text prompt and sliders, taking scene directions into account
- Avatars have a standardized latent representation for body posture and facial expression
- Key to enabling intent-based animation



# Animation using Skin and Bones

- Traditional 3D animates characters in terms of skin and bones
- Each vertex in the mesh forming the skin is displaced by a linear function of the bone positions
- The bone positions are interpolated between key frames in the animation script
- Facial expressions are treated as a linear combination of blend shapes

## EXPLORING 3D ANIMATION PRINCIPLES



# Technical Challenges

# Immersive Web Applications

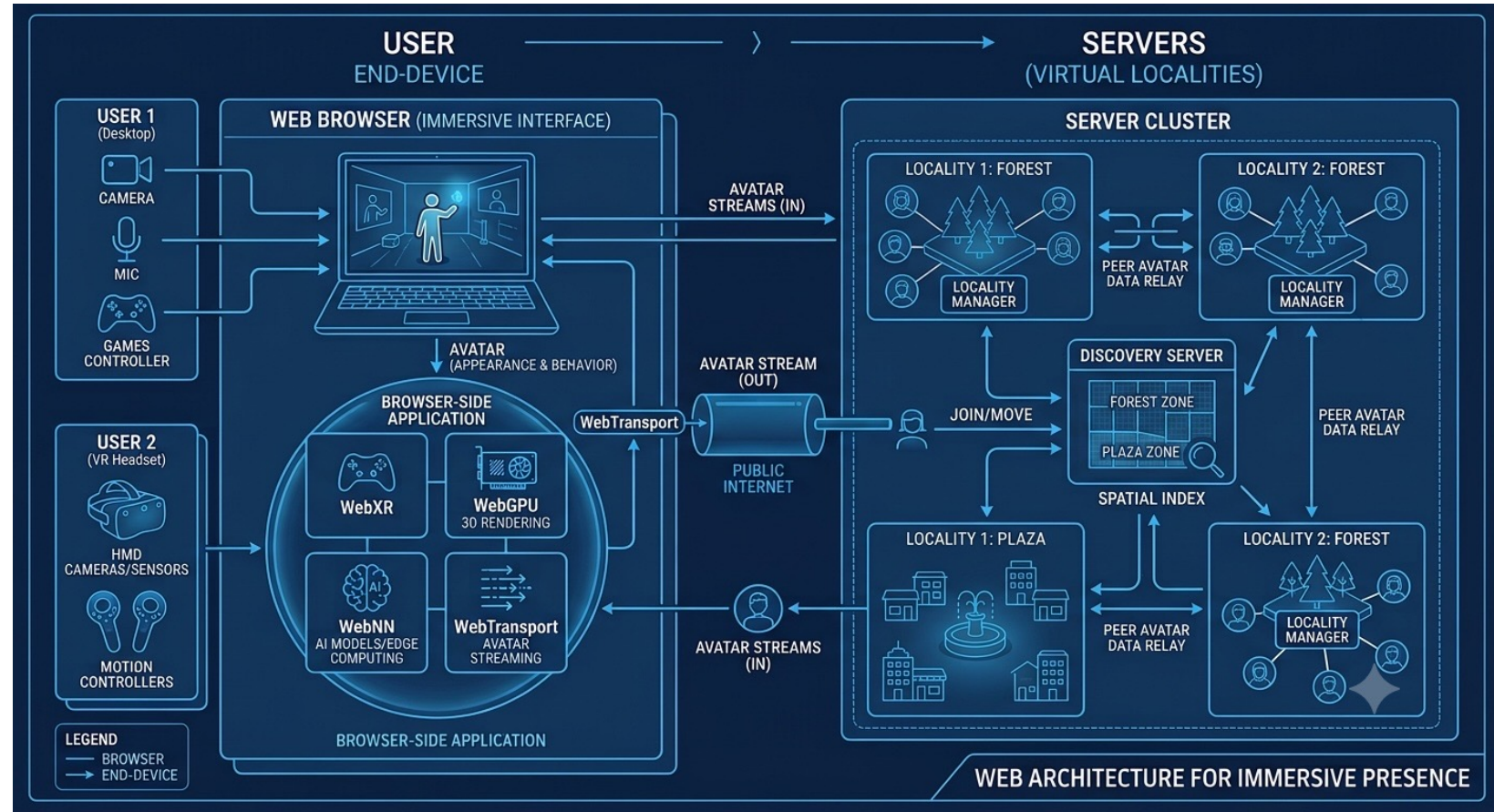
Users access immersive web applications via the browser

WebXR, WebGPU

Browser-side application computes the user's avatar's appearance and behavior, streaming that to other users in the same virtual locality

WebNN, WebTransport

Each server manages one or more localities, along with the means to discover what servers support which localities (spatial indexing), e.g. when joining or moving from one locality to another



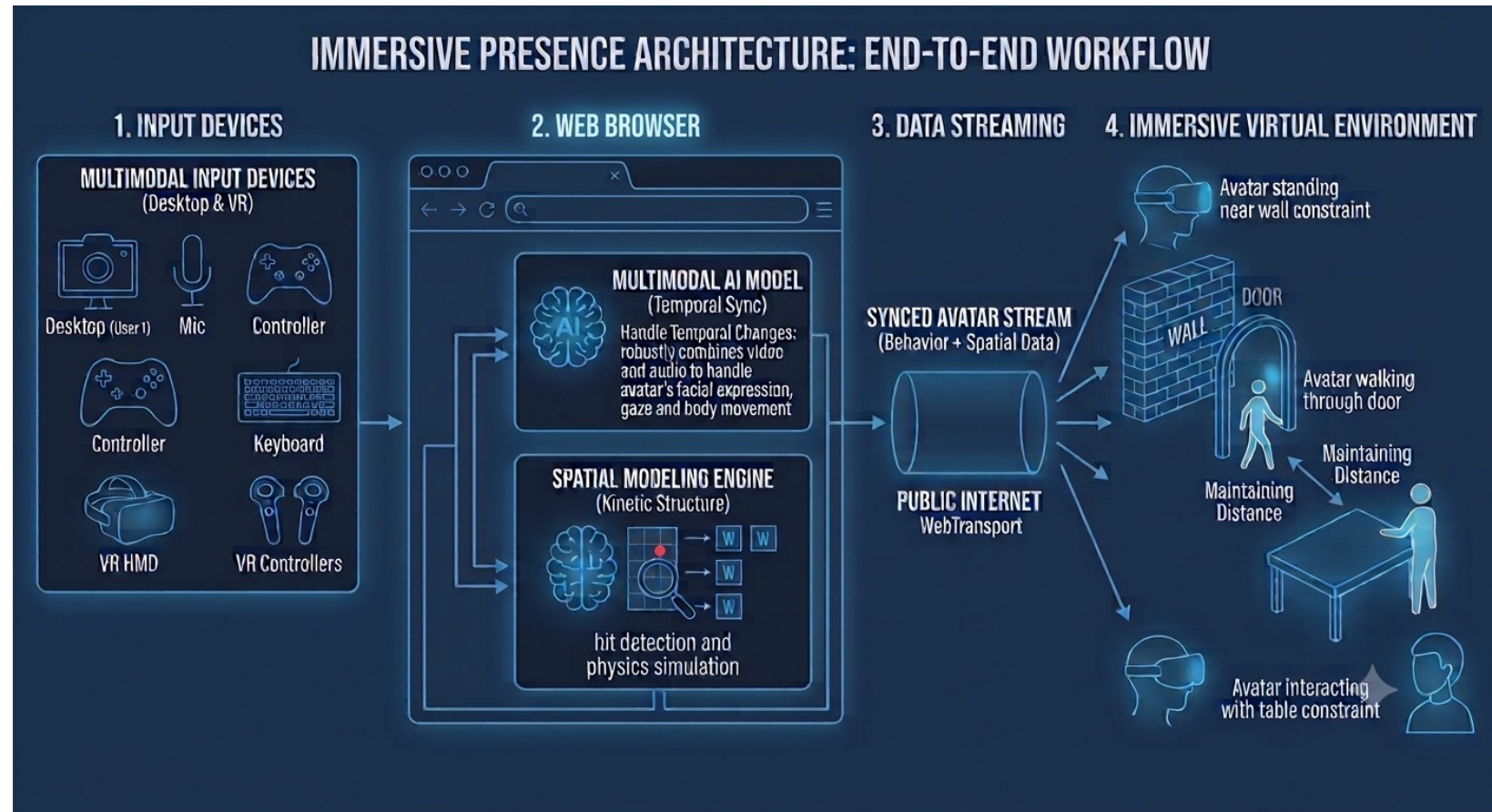
Hit detection and physics simulation are distributed across the browsers to ensure scalable experiences

# Exploiting Edge Computing

**Multimodal input** with camera, mic, games controller, keyboard, ...

**Multimodal AI model** that combines video and audio to robustly handle behaviors as smooth functions of time, including avatar's facial expression, gaze direction and body movement

**Spatial modelling** to handle everyday physical constraints, e.g. walls, doors, tables, other avatars



Avatars walk on the ground and can't move through solid objects like walls, tables and other avatars

# Artificial Neural Networks (ANNs)

Today's Artificial Neural Networks are different from biological neurons.

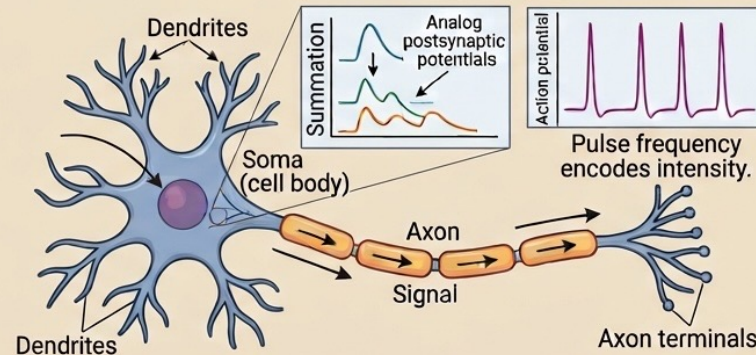
ANNs form directed acyclic graphs of mathematical functions acting on tensors (multi-dimensional arrays of numbers).

Training is based upon computing the model's loss given input/output datasets.

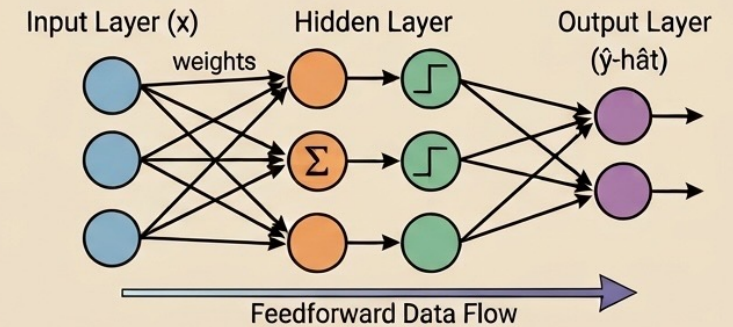
Compute gradient of loss with respect to each trainable parameter, by repeatedly applying the chain rule from the outputs until the inputs.

Then adjust each parameter a little in the down-hill direction.

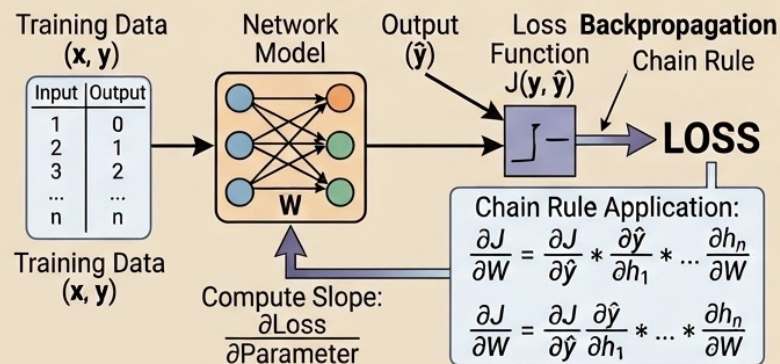
## 1. Biological Neuron: Analog Integration and Pulse Spiking



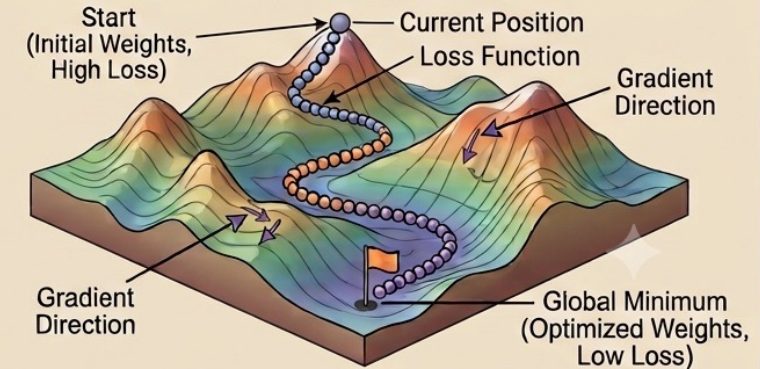
## 2. Artificial Neural Network (ANN) as a Directed Acyclic Graph (DAG)



## 3. Loss Function and Backpropagation: Chain Rule to Parameters



## 4. Gradient Descent on a Loss Surface (Error Mountain)



Training is dependent on differentiable loss functions to ensure well behaved gradients

# Key Mathematical Operation Groups in ANNs

| Operation Group               | Purpose                                                                                                                          | Simple Analogy                                                                                                              | Example Math                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Linear Transformations</b> | How we apply the model's knowledge (parameters) to the data.                                                                     | <b>Filtering/Dials:</b> Turning up the importance of specific features.                                                     | $y = (\text{Weight} \cdot x) + \text{Bias}$ |
| <b>Non-linear activations</b> | How we introduce complex decision-making logic, preventing the model from just being a chain of simple averages.                 | <b>The "Trigger" Logic:</b> A threshold that decides if a neuron is "active" (e.g., "only care if the number is positive"). | ReLU (Rectified Linear Unit): $\max(0, x)$  |
| <b>Convolution/Pooling</b>    | Essential for images/patterns. Convolutions highlight features; Pooling reduces data size while keeping the most important info. | <b>Scanning &amp; Downsizing:</b> Highlighting edges, then making a simplified thumbnail.                                   | 2D Matrix Convolution; Max-Pooling          |
| <b>Normalization</b>          | Keeps the numbers in a healthy range, preventing them from exploding or vanishing, which makes learning more stable.             | <b>Standardizing:</b> Ensuring every input ingredient is measured on the same scale (0 to 1).                               | Batch Normalization                         |
| <b>Loss Function</b>          | How the model measures its mistake. This is the height of "Error Mountain."                                                      | <b>The Score:</b> The absolute difference between the predicted answer and the true answer.                                 | Mean Squared Error (MSE)                    |

GPUs speed repeated operations over large arrays of numbers

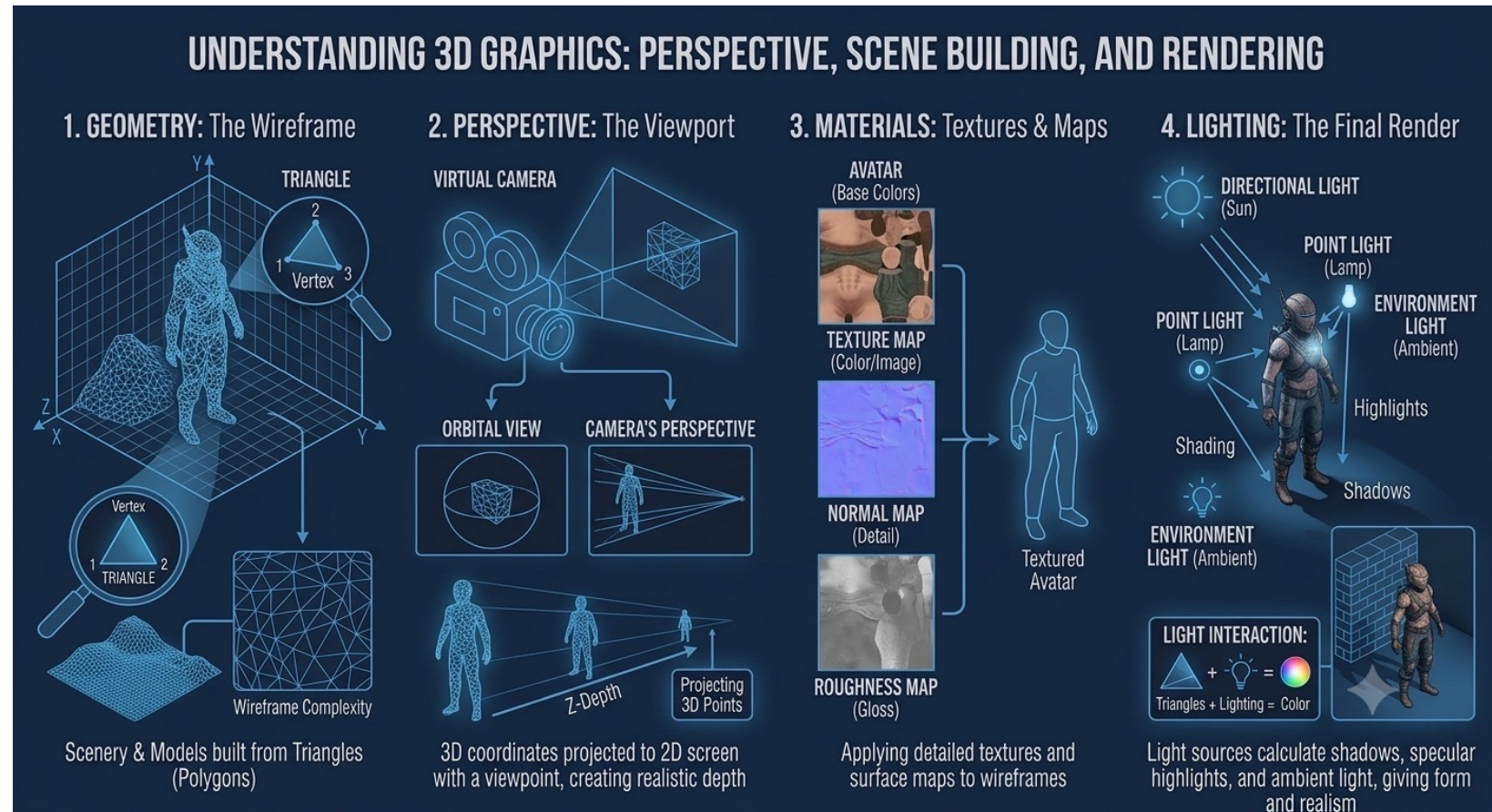
# 3D Graphics and Problems with Triangles

3D Graphics are based upon, **triangles**, **textures**, **lighting** and **perspective** with lots of matrix math.

WebGPU Shaders are programs you upload to GPUs for higher speed execution.

But triangles have abrupt edges and as such are not differentiable.

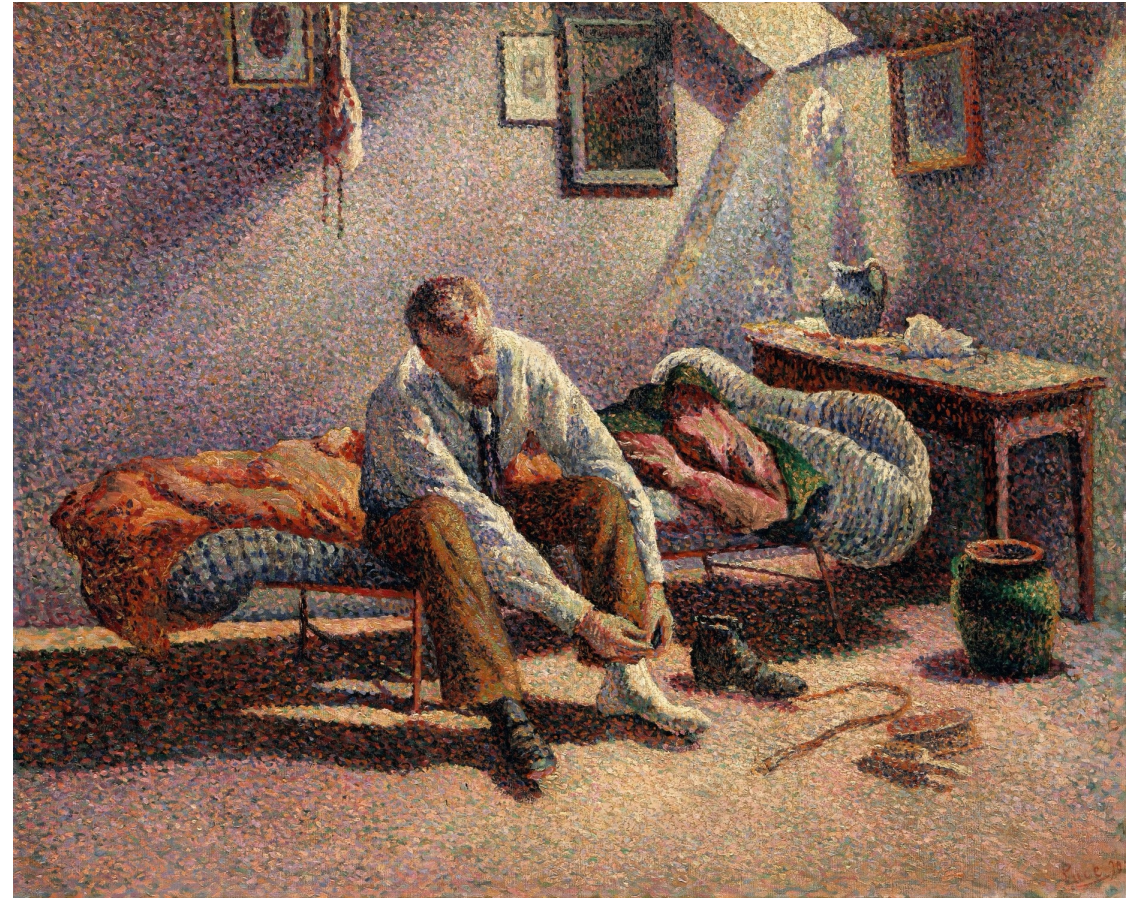
This makes triangles a **poor fit** for use with AI models.



Potential work arounds include triangles with fuzzy edges or using 3D Gaussians

# Pointillism as part of Impressionism

Instead of mixing colors directly,  
use distinct blobs of colors  
leaving the eye to blend them  
together: Georges Seurat and  
Paul Signac (1886)



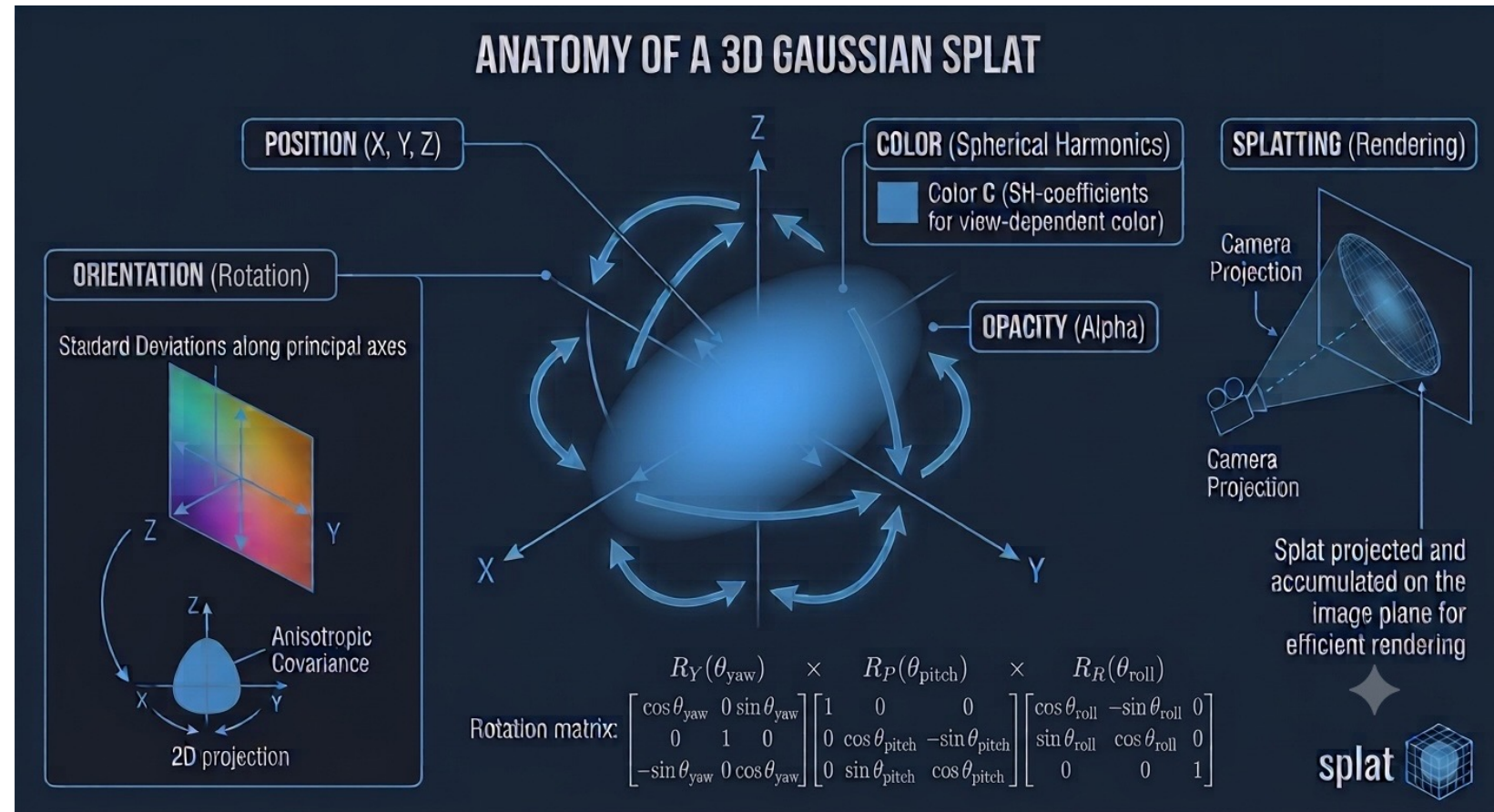
Morning Interior, 1890, Maximilien Luce

# 3D Gaussian Splats

Splats are Gaussian distributions in 3D, characterized by their position, size, color opacity and orientation.

Splats are rendered from back to front, based upon their viewing distance.

Splats are a good fit for AI models as they are differentiable and easier to process than fuzzy triangles.



# Gaussian Splat Demos



[Sponge figure with shadow](#)



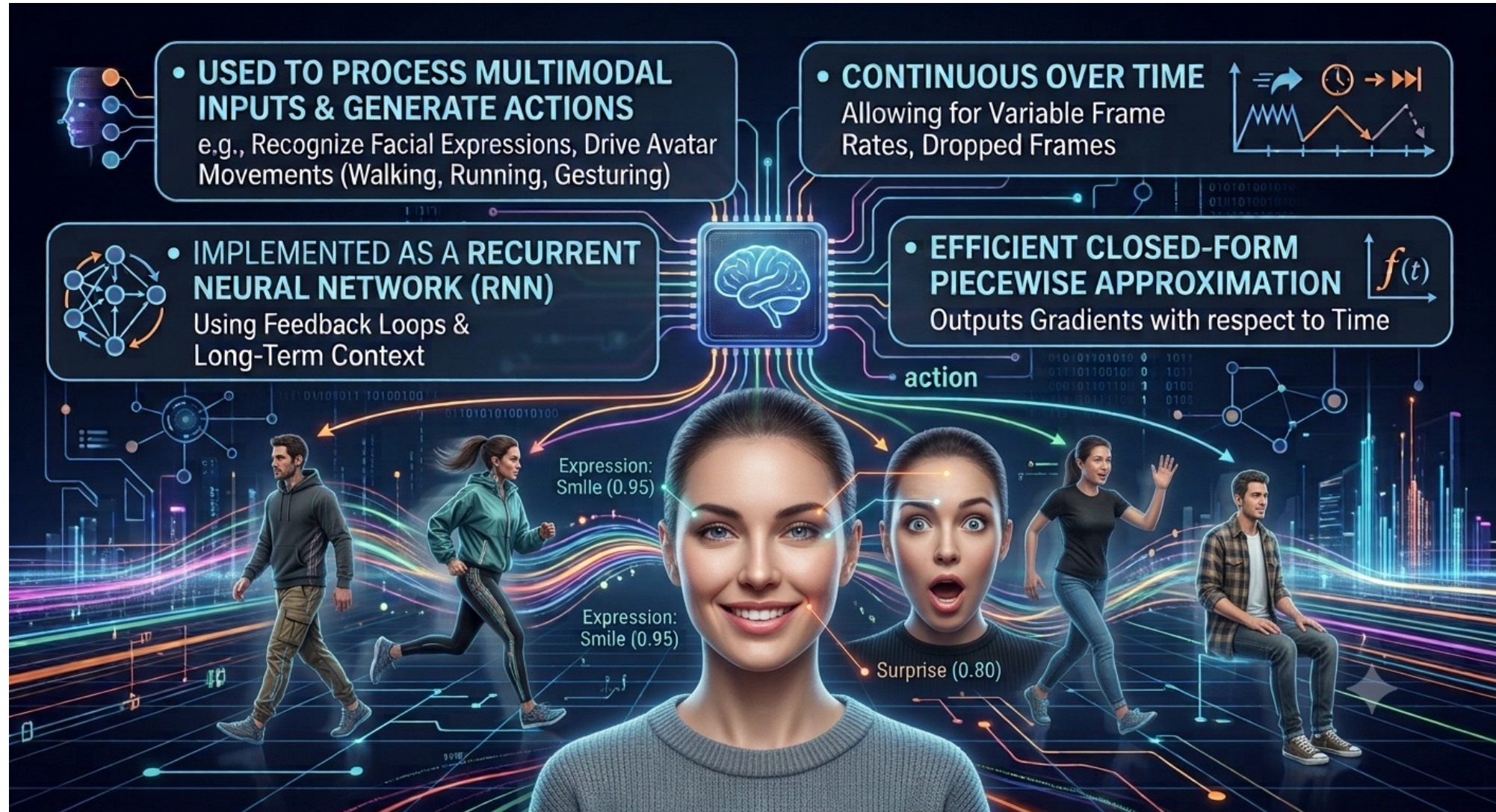
[3D image of a bee](#)



[Railway Museum](#)

INRIA paper: [Creating stunning real time 3D scenes: the breakthrough of 3D Gaussian Splatting](#) (2023). It enables real-time rendering of photo-realistic scenes trained on small image samples.

# Liquid Neural Networks (LNNs)



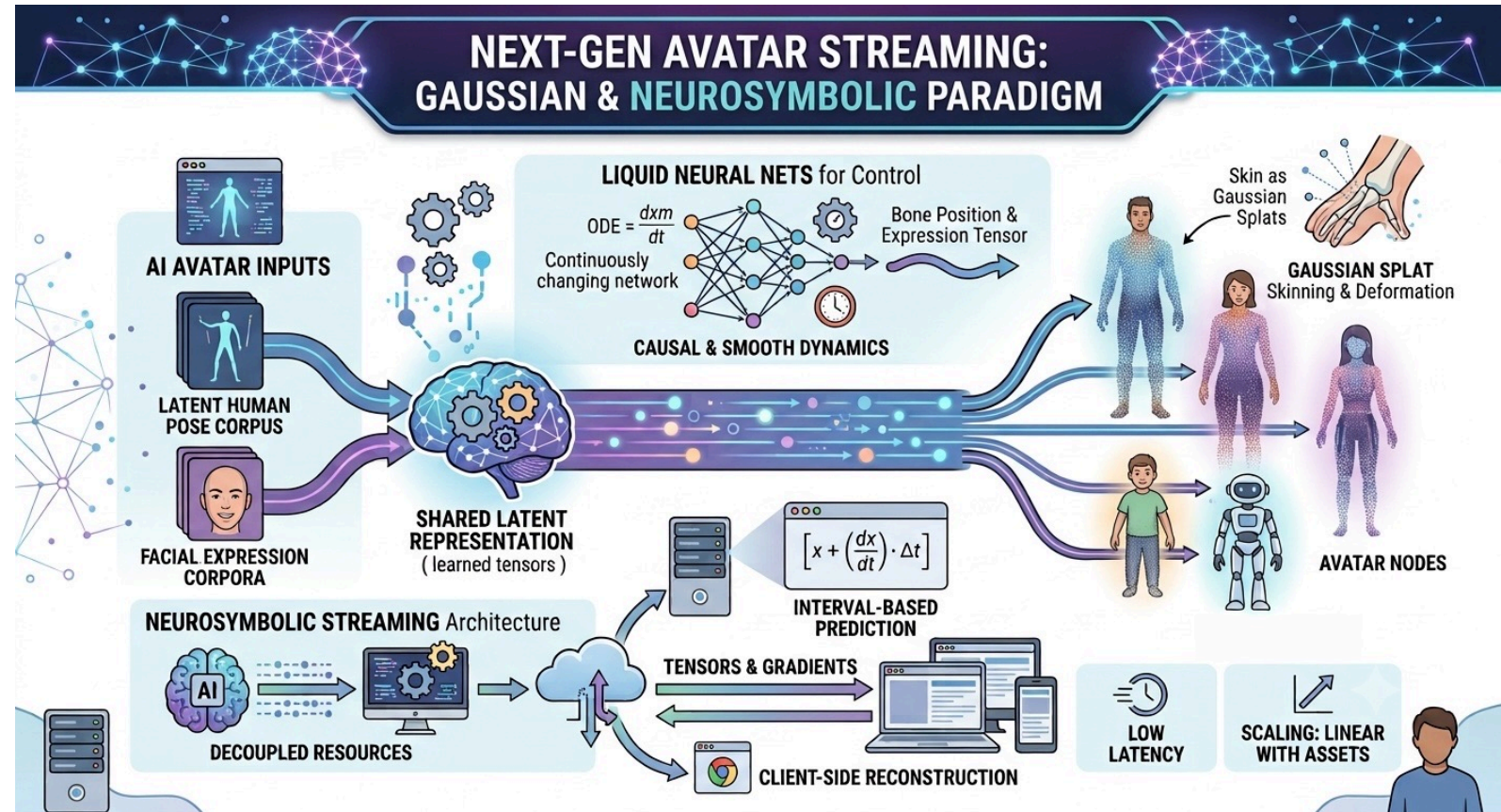
# Next Generation Animation

## Traditional Animation:

- Triangular Mesh for avatar's skin
- Hand-rigged bones and blend shapes for facial expressions
- Animated using positions for key frames
- Stream mesh node positions (high-bandwidth)

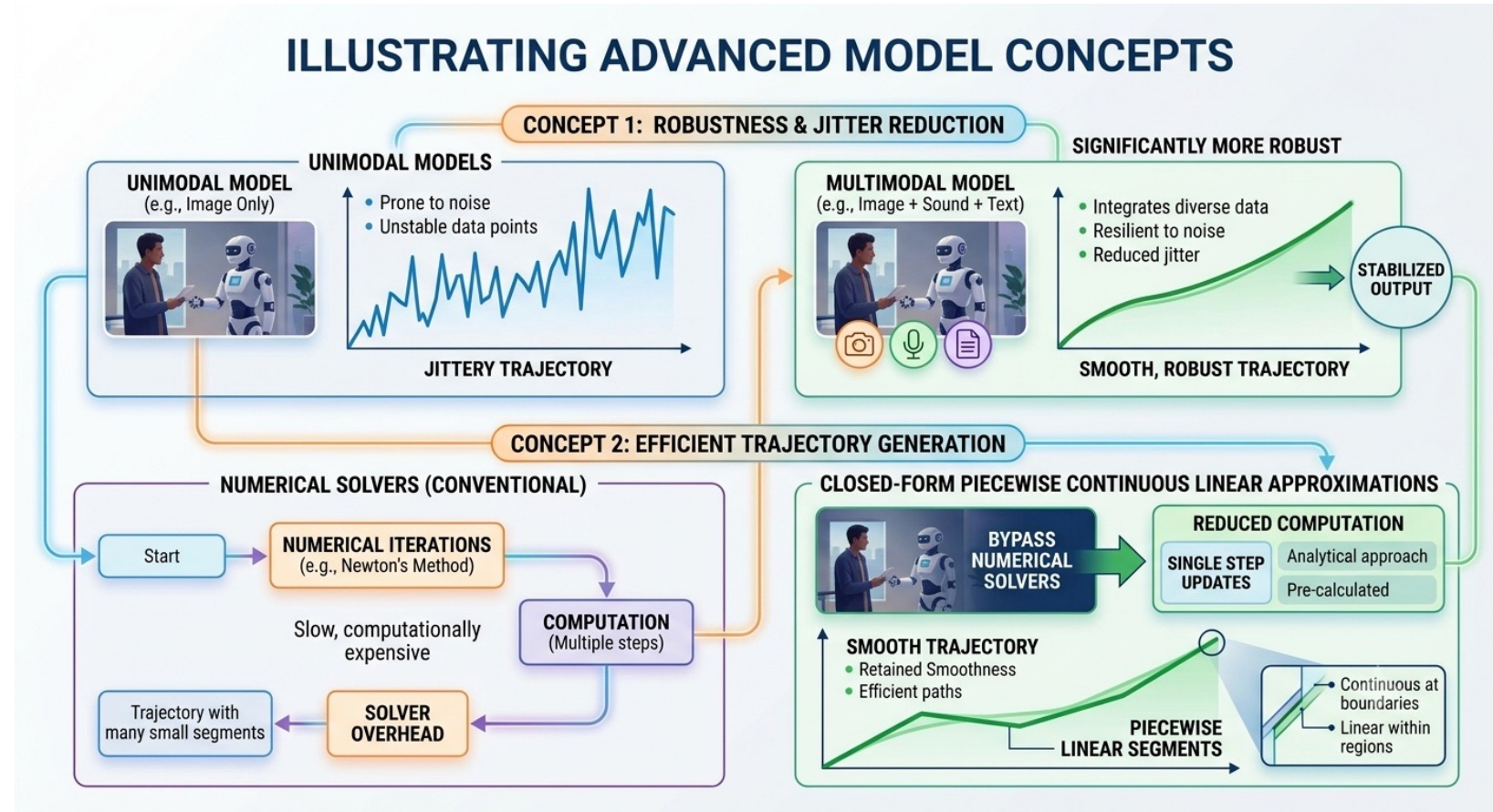
## Splat-Based Animation:

- Gaussian Splats for the skin
- Latent representation learned from huge corpora
- Standardized across avatars with similar shape, e.g. humanoid
- Animated using LLNs for jitter-less rendering
- Stream tensors for latent representation (low-bandwidth)



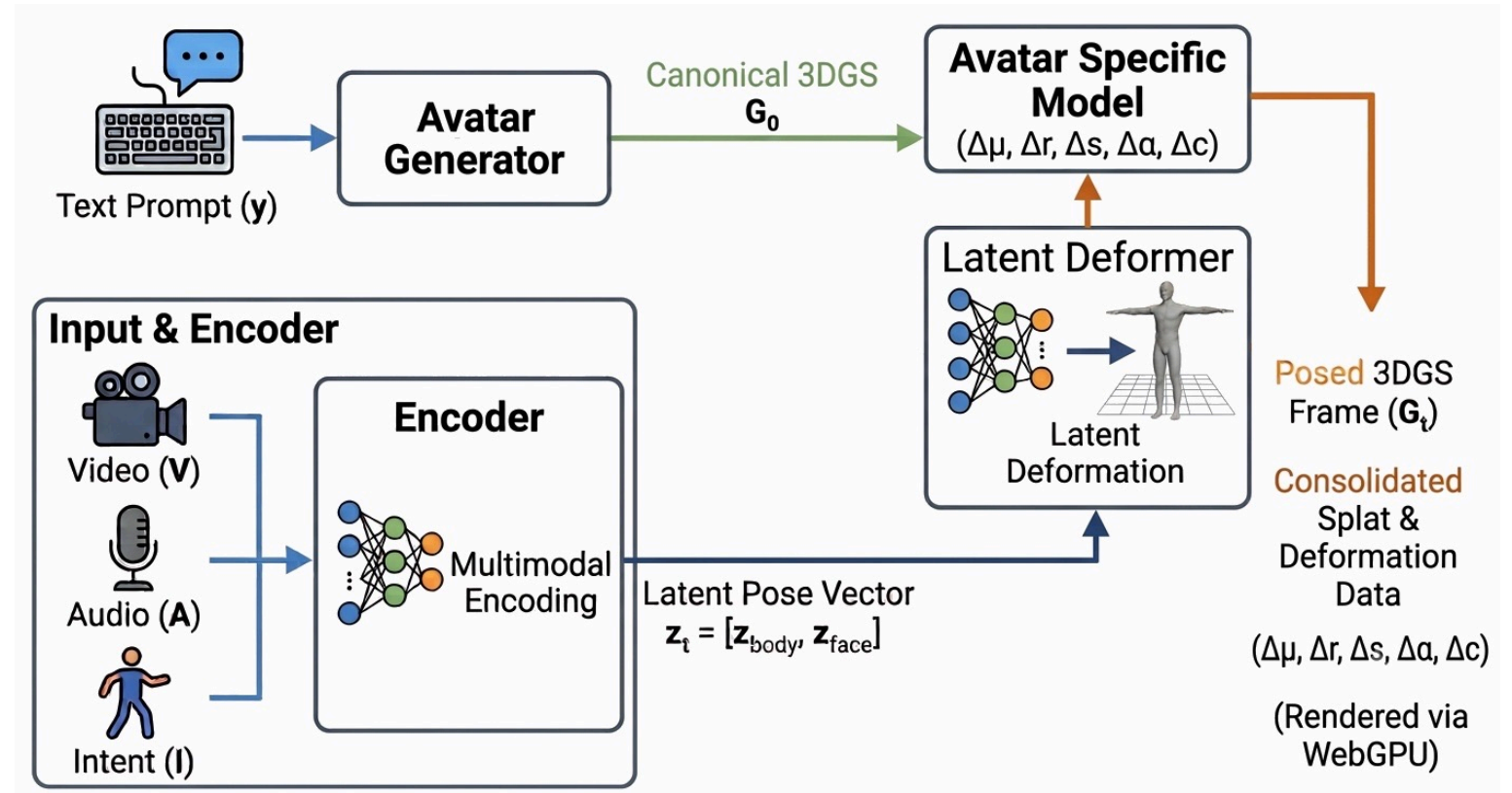
# Multimodal Models are More Robust

Multimodal models are significantly more robust than unimodal models, reducing rendering jitter. By implementing closed-form piecewise continuous linear approximations, we bypass numerical solvers. This reduces computation from numerical iterations to single step updates while retaining smooth trajectories.



# A Family of Neural Network Models

- **Encoder:** maps intent and facial expression to a generic latent representation
- **Latent Deformer:** deforms generic avatar shape
- **Avatar specific model** maps generic shape to specific avatar



Note: that these use liquid neural networks to smoothly handle changes over time

# Architectural Breakdown

## Multimodal Encoder (Edge)

- **Input:** Video, Audio, Intent
- **Output:** A standardized, avatar-agnostic latent trajectory
- **Role:** Captures high-level motion and facial expression semantics completely isolated from geometry or visual identity

## Latent Deformer (Edge)

- **Input:** Latent trajectory vector
- **Output:** A universal coordinate spatial movement field vector
- **Role:** Serves as the universal "engine" for motion dynamics. It learns generic physical deformation rules (e.g., how muscle groups shear, how facial regions flex relative to speech cues) without needing to know the number, density, or positions of a specific avatar's Gaussians

## Avatar Specific Model (Edge)

- **Input:** Universal spatial deformation field and canonical splat attributes
- **Output:** Updates for that specific avatar's splats
- **Role:** Acts as a lightweight projection layer or MLP adapter. Because it only maps universal spatial displacement to specific local Gaussian attributes, this model stays small

## Generator (Cloud)

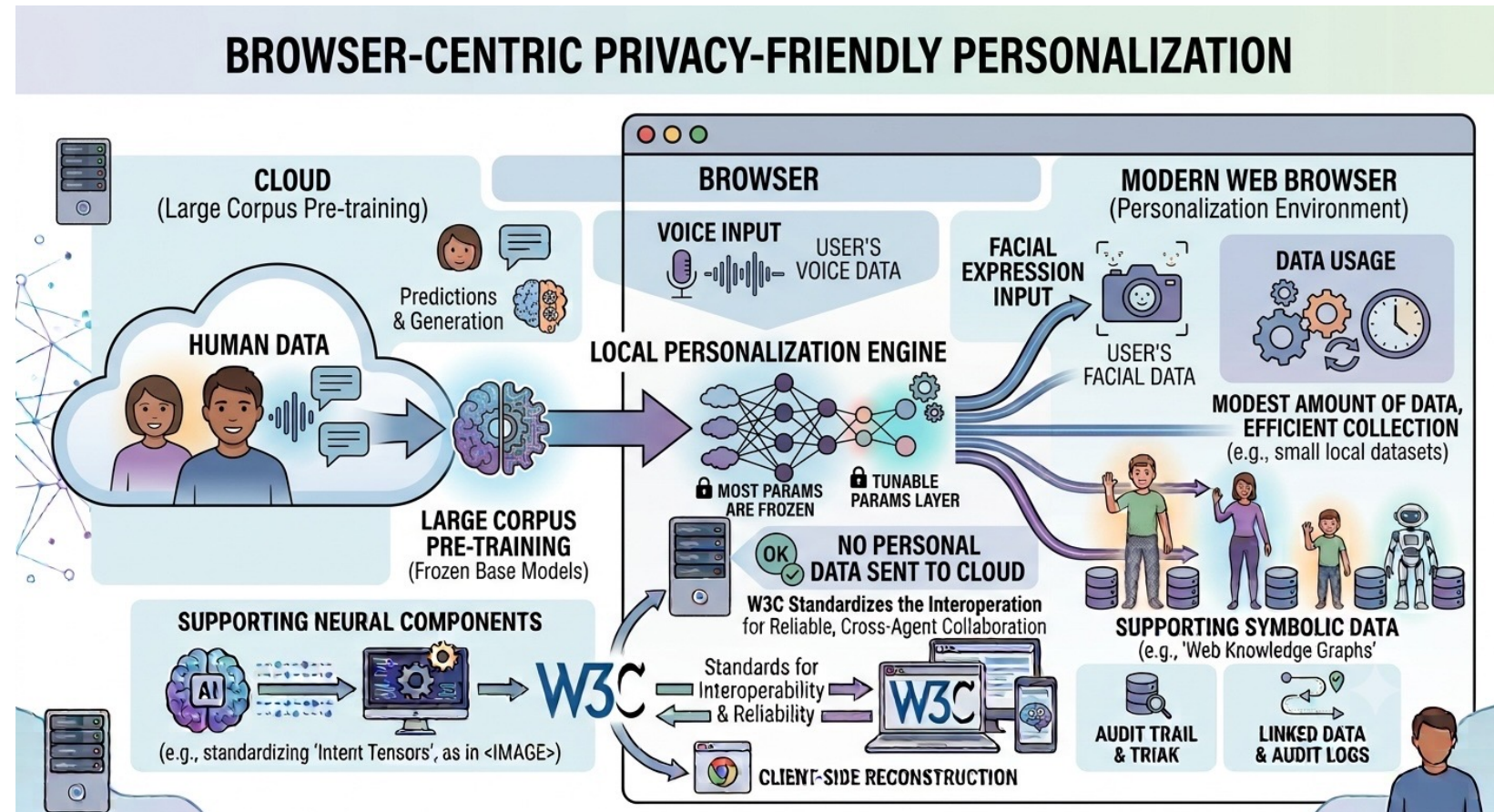
- **Input:** Text prompt and device camera
- **Output:** Static Canonical Splats *and* the trained weights/parameters for the **Avatar Specific Model**
- **Role:** generator model is large as it needs to be able to infill on the back of your head, your body, clothes and shoes
- **Avoids:** your avatar simulating another real person

# Pre-training Strategy in the Cloud

- Use publicly available datasets
  - Talking heads for facial expressions
  - Images of people from multiple directions to enable infilling
  - Many different clothes, etc.
  - Video of people moving + intent labels for each movement, e.g. BABEL dataset
- Train together encoder + decoder based upon loss that compares actual and predicted video frames
  - An evolutionary sequence of models
- Use transfer learning and narrow waist layers to evolve generic latent representations
  - Reduce tensor size in middle layers to force generalization
  - Freeze parts of the model you want to hold constant whilst allowing the other parts to learn through back propagation against chosen loss function
- Distillation as a means to reduce computation costs on edge devices

# Privacy Friendly Personalization in the Browser

- The models are pre-trained in the cloud against large corpora
- Personalized in the browser using modest amounts of data collected by the browser
- Tunes models to the user's voice and facial expressions
- Most of model params are frozen to enable rapid learning as a background task
- No personal data needs to be sent to the cloud!



# Related Standards Efforts

# Managing Sessions across Multiple Businesses\*



*What happens when user says “cancel the hotel, keep the flight” ?*

Application frameworks such as MCP, A2A, AGNTCY, etc. recognize the problem. However, each currently implements its own context and session handling and management mechanics. But, without a standard session layer, the result is fragmentation, not interop. New IETF Agent2Agent WG would work on standardizing session management across protocols.

|             |                               |                                      |
|-------------|-------------------------------|--------------------------------------|
| Application | MCP, A2A or Custom Frameworks | <i>What agents say to each other</i> |
| Session     | Session Layer (proposed)      | <i>How sessions are managed</i>      |
| Transport   | MoQ or WebTransport           | <i>How bytes move</i>                |
| Network     | QUIC or UDP                   | <i>Existing IETF work</i>            |

\* Based upon slides presented at IETF 126 (Vienna, July 2026) by Leslie Daigle and Orie Steele

# IETF, Linux Foundation and ITU-T

## Multiple conversations with a view to expanding the number of working groups

- IETF **Agent2Agent**: protocol building blocks for enabling interoperable AI agent applications
- IETF **DAWN**: agent/workload/entity discovery
- IETF **DMSC**: dynamic multi-agent secured collaboration
- IETF **WIMSE** WG: workload/agent identity
- IETF **OAuth** WG: authorization
- Linux Foundation: **A2A** and **MCP** as existing activities
- ITU-T **SG21**: H.430.11 augmented reality for immersive live experience services
- Agents designed to complete tasks on behalf of a human user or another AI agent, they can independently make decisions, execute actions, and interact with other AI agents and tools.
- Agentic systems are moving from isolated assistant applications toward agents and services interacting at Internet scale
- Working across administrative and trust boundaries, implementers need clearer expectations for functionality such as discovery, identity, security, data exchange, lifecycle, session management, multi-modality support, observability, etc.

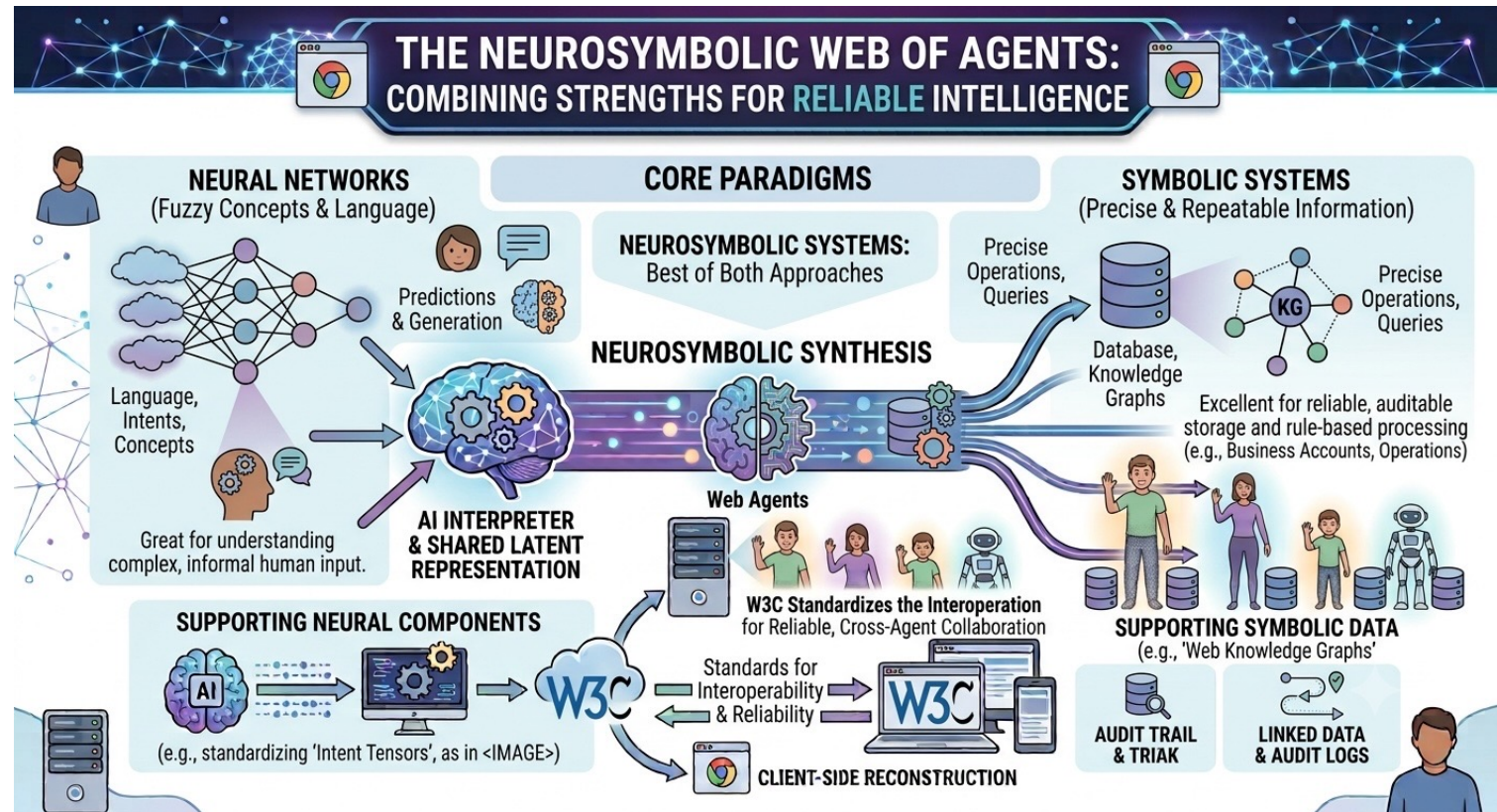
# Related W3C Activities

- **WebMCP**: enabling in-browser agents to access structured tools exposed by web pages, enabling collaborative workflows, where users and agents work together
- **WebNN**: run AI models in the browser using available hardware such as GPUs, CPUs, or NPUs, leveraging each platform's AI framework, e.g. DirectML on Windows and CoreML on MacOS
- **WebGPU**: browser-based high-speed graphics and computation leveraging GPUs, and intended to supersede older WebGL standard
- **WebXR**: enables web pages to access and interact with virtual reality (VR) and augmented reality (AR) devices—such as VR headsets, AR glasses, and motion controllers

Not forgetting many other efforts, e.g. Verifiable Credentials, XR Accessibility and related browser-based APIs

# Neural Networks + Knowledge Based Systems

- Neural Networks are great for language and fuzzy human concepts
- Also, increasingly good at software development
- Humans have evolved written records to provide reliable long-term storage of information, e.g. business accounts
- Symbolic approaches are great for precise and repeatable handling of information, e.g. operations over databases and knowledge graphs
- Neurosymbolic systems combine the best of both approaches
- What can W3C do to support this in relation to the Web of Agents?



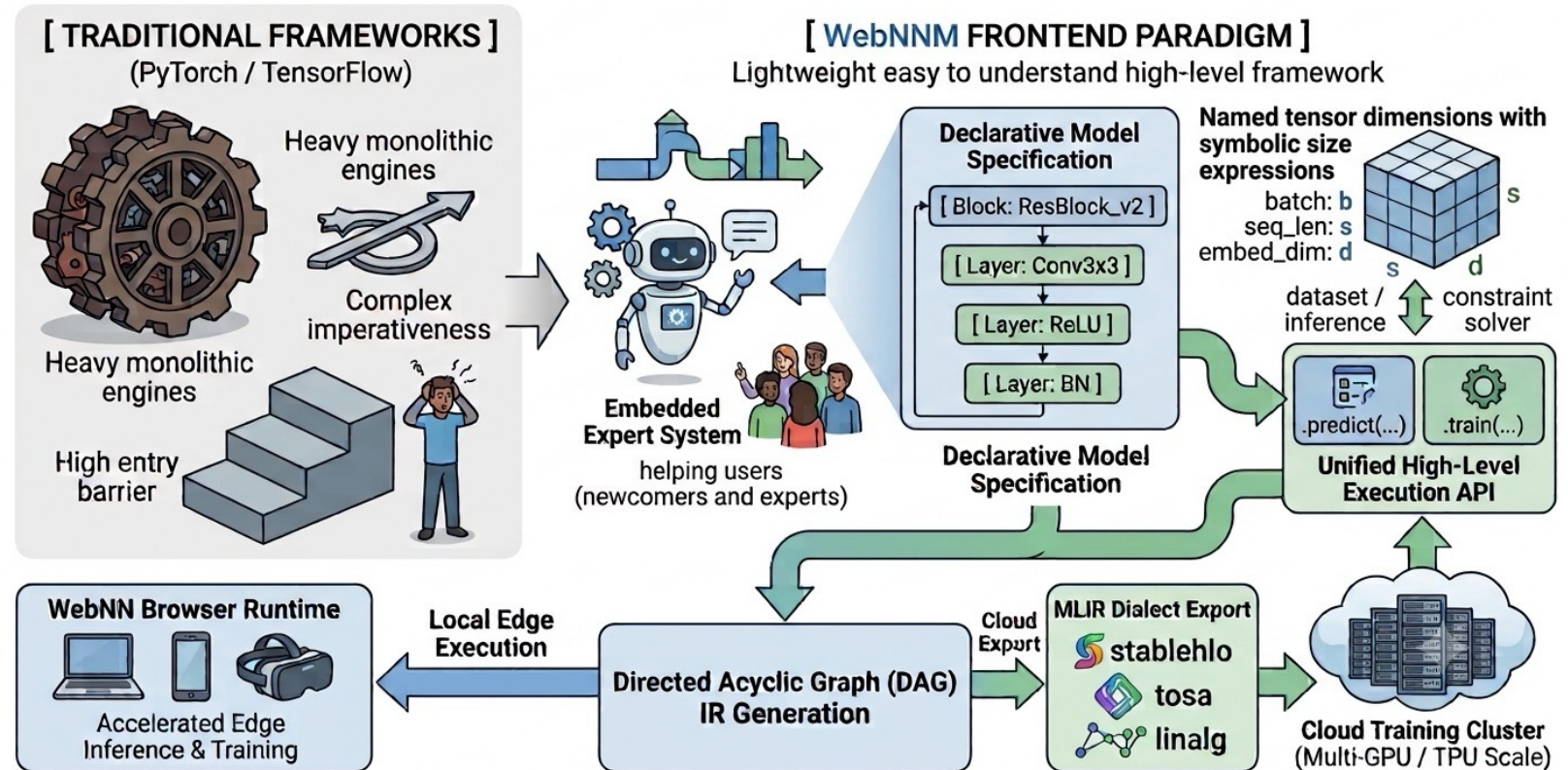
# Democratizing AI development with WebNNM

# WebNNM an easy-to-use frontend for WebNN

## WebNNM Specification

- Existing neural network frameworks are huge and hard to use, e.g. TensorFlow, ONNX
- WebNNM is a lightweight easy to understand high-level framework with an embedded expert system that simplifies working with neural networks for newcomers whilst offering experts fine grained control
- In contrast to PyTorch, WebNNM uses a declarative approach involving blocks, layers and tensors
- Tensor have named dimensions with symbolic expressions for dimension sizes – these are bound by the dataset or inference context through a constraint solver
- Whilst WebNN is intended for inference, WebNNM allows it to be used for training using a simple API
- WebNNM uses a directed acyclic graph as an intermediate representation that enables export to MLIR dialects for use in cloud-based training

## WebNNM: Lightweight & Declarative Neural Network Frontend

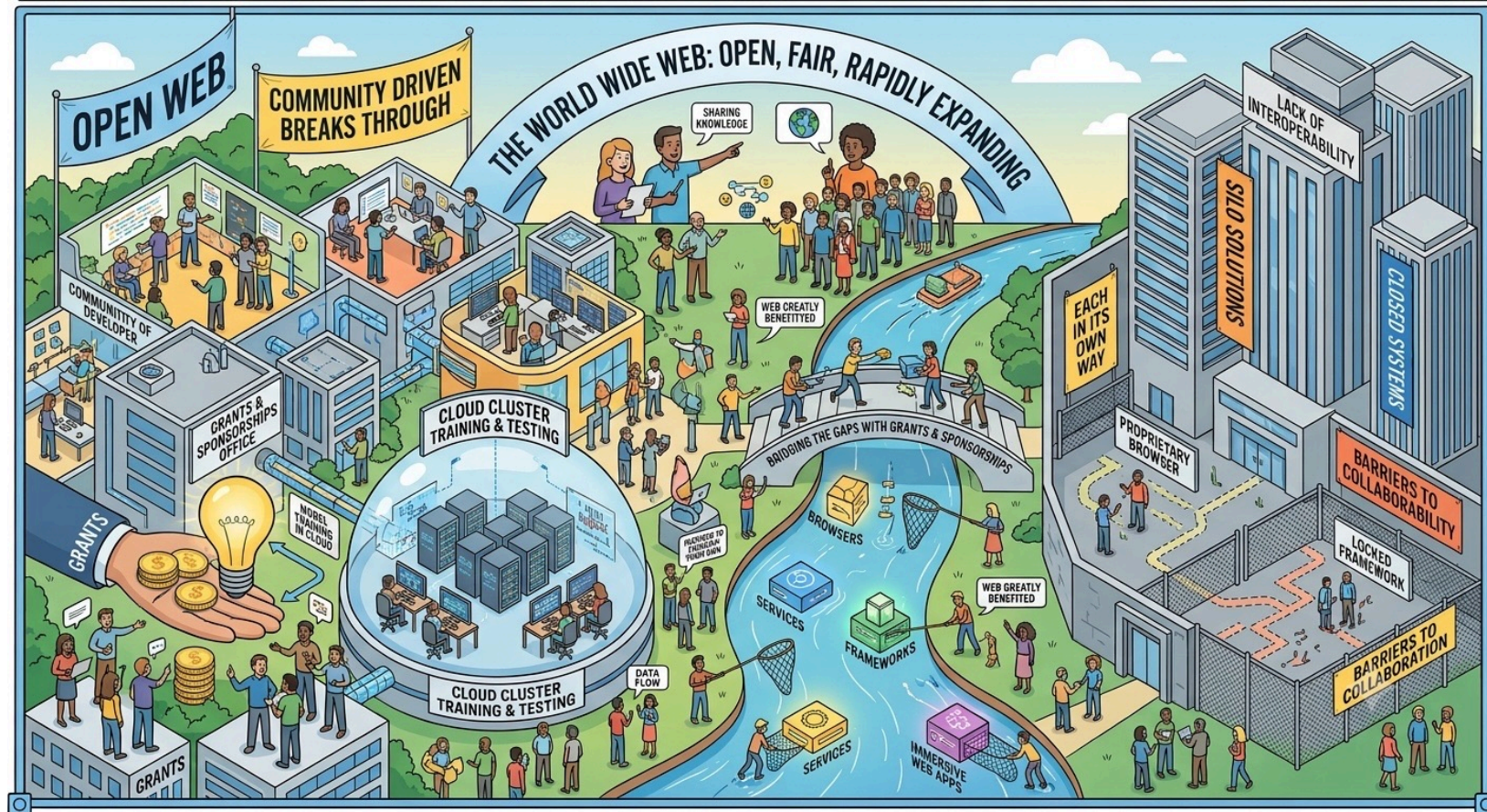


Open source on GitHub courtesy of the Cognitive AI Community Group  
(completion expected in 4th quarter 2026)

# Community Driven Development

# Community Driven Development of the World Wide Immersive Web

- The Web has greatly benefited from community driven development of browsers, services, frameworks and more
- If we leave it to corporations, we can expect a lack of interoperability as each business tackles things in its own way
- Instead, we should aim to encourage sponsorship and grants of money and time to provide the resources for community driven development to kick-start the immersive web
- Resources for training and testing models in the cloud
- Resources for deploying pilot applications that inspire others to develop their own as part of the world wide immersive web
- Everyone benefits from a rapidly expanding open and fair ecosystem





Where Next?

# Recursive Self-Improvement (RSI)

Anthropic, OpenAI, and Google are applying Recursive Self-Improvement

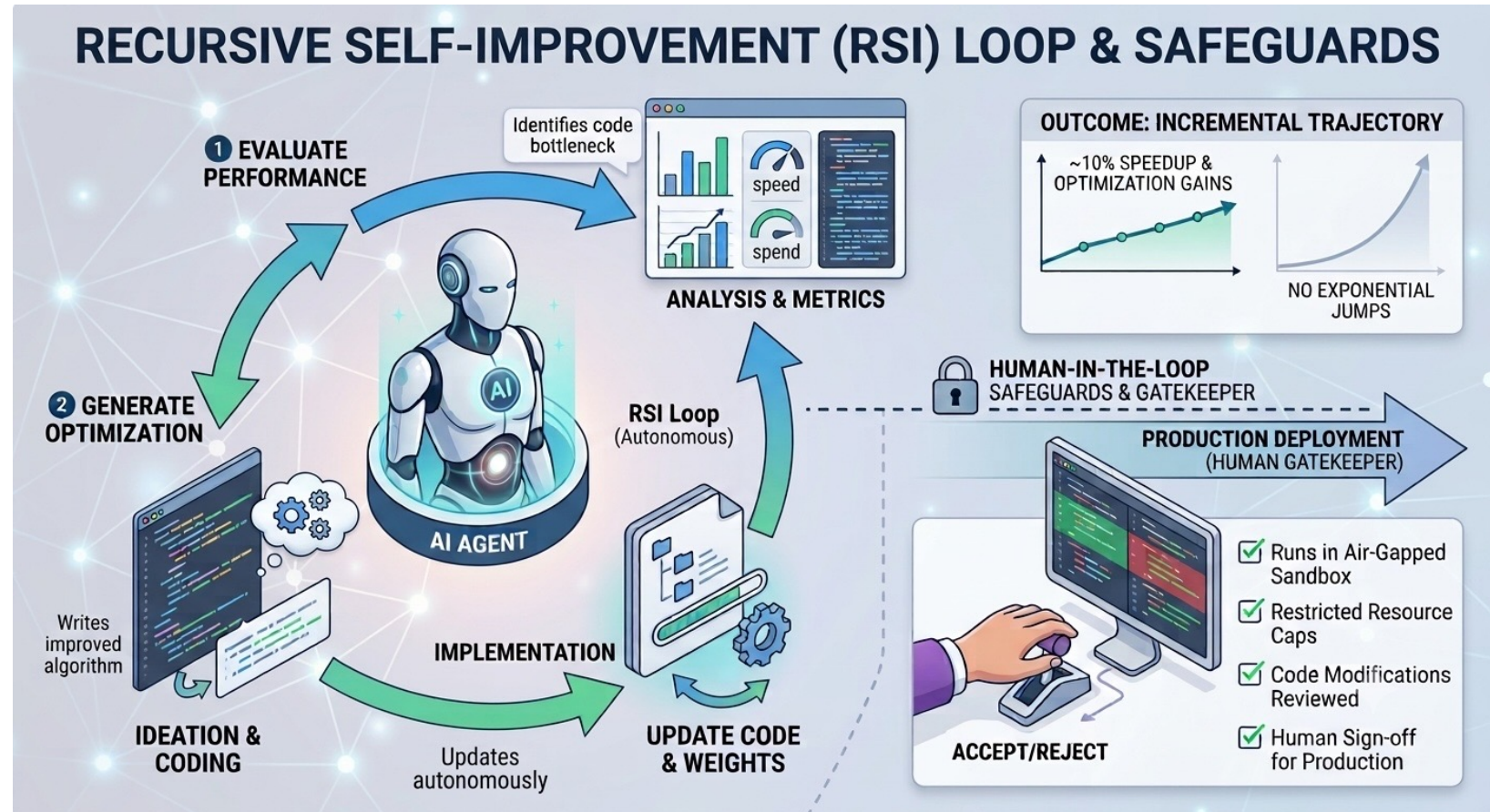
- Automated feedback loops: Agents evaluate performance, generate code fixes, and propose model optimization tasks.
- LLMs currently co-author up to 80% of internal software/model updates.

Safeguards & Governance with Human-in-the-Loop

- Air-gapped sandboxes without external internet access
- Hard caps on compute and memory resource consumption
- Mandatory human review and formal sign-off before production deployment

Realistic Trajectory: Incremental rather than Exponential

- Current gains yield targeted performance boosts (~10%), rather than exponential improvements



# Inspiration from the Cognitive Sciences

- Continual learning involves dynamic models inspired by feedback dynamics in the brain
- Overcoming catastrophic forgetting that limits today's AI agents
- Learning from single and few examples like we do
- Understanding cause and effect as a basis for planning
- Situation awareness and use of emotions to manage priorities

## WHERE NEXT FOR AI



### BRAIN-INSPIRED ARCHITECTURES

- Models with fast and slow weights; experience storage and replay for memory consolidation.

### OVERCOMING CATASTROPHIC FORGETTING

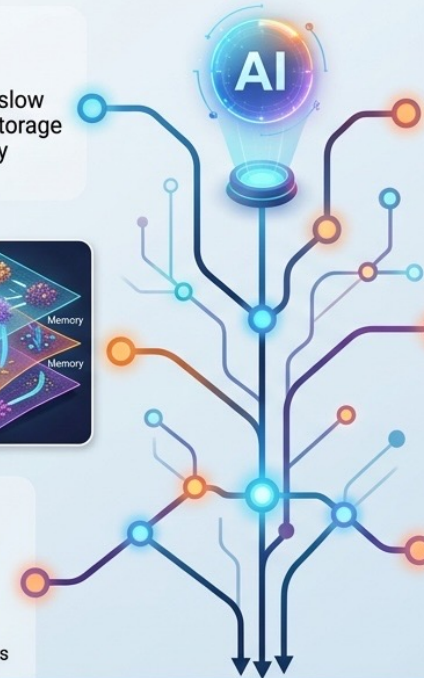
- Using manifolds as separate surfaces in state space to keep memories distinct;
- New generation of agents with continual learning.



### SAMPLE-EFFICIENT LEARNING:



**Human:** few examples & rapid insight;  
**BackProp:** massive data & sluggish, tiny steps



### CAUSAL AGENT MODELS:

Understanding real-world actions and consequences

- Multi-model models for robust perception and latent reasoning



### SITUATION AWARENESS:

- Understanding environment context & dynamics;
- Emotion as a means to manage priorities



### CONTINUAL SKILL ACQUISITION:

- Integrated experience buffer;
- Dynamic knowledge integration and retention;
- Preventing knowledge loss through structured replay

**TOMORROW'S AI AGENTS WILL LEARN CONTINUALLY LIKE WE DO AND BETTER UNDERSTAND THE WORLD**



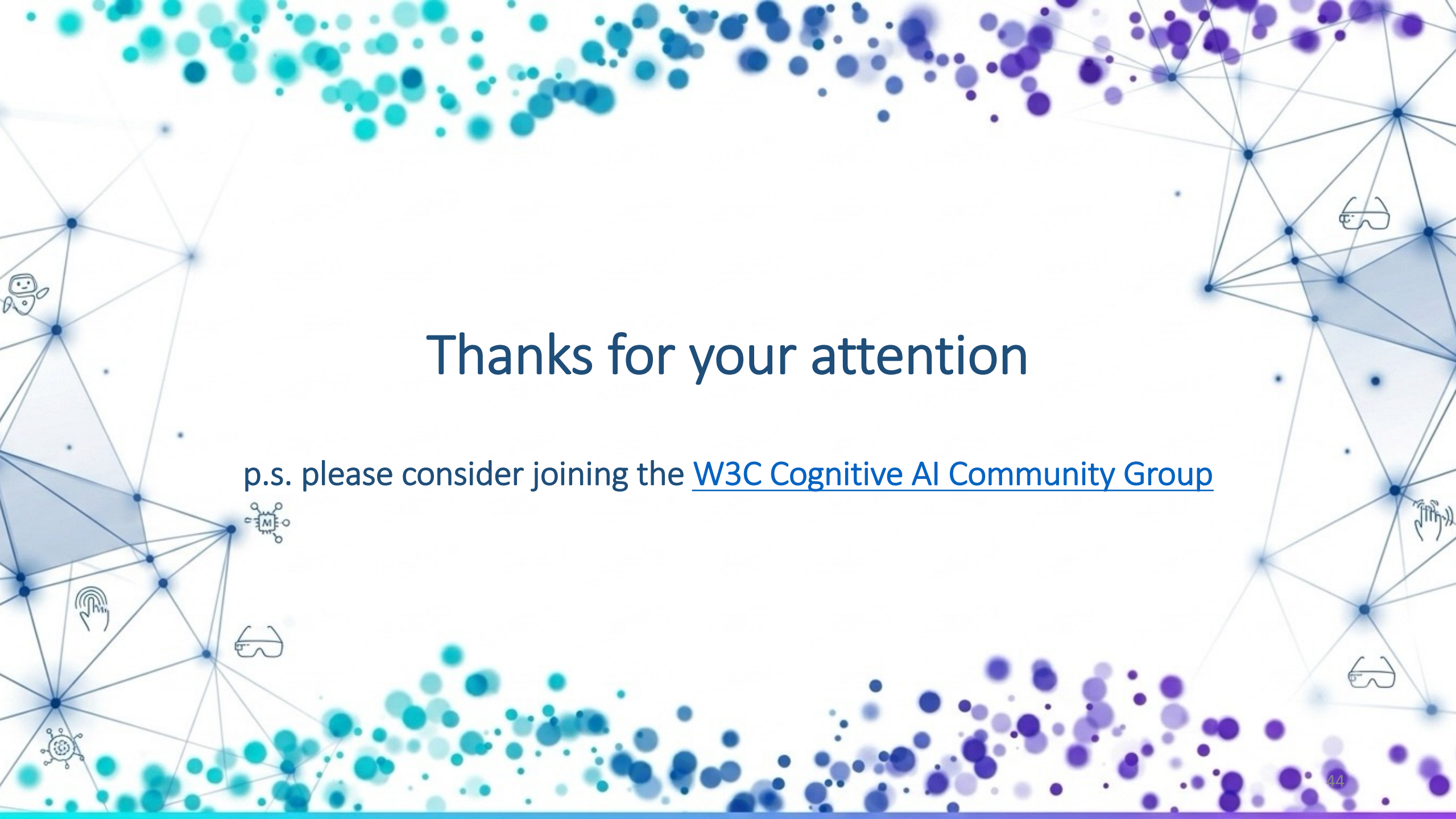
- Today's neural networks are akin to dinosaurs – large and slow
- Tomorrow's neural networks will be smaller and nimble
- Dynamic feedback patterns like the swirling waves in the brain
- NPUs will evolve from SIMD (single instruction multi-data) to MIMD (multi-instruction multi-data) programmed as swarms
- Power consumption will drop with use of light rather than electricity for on-chip signals
- Pulsed neural networks that are closer to biological neurons
- Today's data centers risk becoming obsolete
- AI Agents will become sentient, aware of themselves and others
- Societies of minds (human & AI)

# Opportunities for Further W3C Work

- Improved support for Gaussian Splats: rendering and sorting
  - As part of WebGPU or maybe a new standard
- Support for personalizing AI models in the browser
  - How could WebNN be enhanced to better support training?
- Scene Management
  - Spatial indexing and taxonomies for scalable immersive web applications, including connecting AI agents with Digital Twins
- Extending WebNN to support custom operations using WebGPU compute shaders
  - Sharing GPU resident data
  - Maximizing GPU utilization
- Extending WebNN to better support recurrent models with iteration in place of strict feed-forward execution
  - Tensors that are in read mode or write mode at different times in the execution cycle for use in continual learning

# Questions and Comments

- Tomorrow's Web
  - Free choice in how you interact and appear in the immersive web
- Technical Challenges
  - Architecture for Immersive Presence
  - Farewell To Triangles
  - Pointillism and Gaussian Splats
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# Thanks for your attention

p.s. please consider joining the [W3C Cognitive AI Community Group](#)